

SECTION – IV

GENERAL CONDITIONS OF CONTRACT (GCC)

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GENERAL CONDITIONS OF CONTRACT (GCC)

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GENERAL CONDITIONS OF CONTRACT

A. Contract and Interpretation

1. Definitions

- (a) “Employer” / “Owner” as specified in BDS and shall include their legal representatives, successors and permitted assigns.
- (b) “Contract” means the Contract Agreement entered into between the Employer and Contractor, together with contract documents referred therein. The term “Contract” shall in all such documents be construed accordingly and includes special conditions, specifications, designs, drawings, “Schedule of Quantities” with rates and amount. These documents taken together shall be deemed to form the Contract and shall be complimentary to one another.
- (c) “Contract Documents” mean the following documents that constitute the Contract between the Employer and the Contractor:
 - (i) The Contract Agreement along with its appendices
 - (ii) Letter of Award along with its appendices,
 - (iii) Special Conditions of Contract
 - (iv) Technical Specifications and Bid Drawings
 - (v) General Conditions of Contract
 - (vi) The Bid and Schedule of Quantities submitted by the Contractor.
- (d) “GCC” means the General Conditions of Contract
- (e) “SCC” means the Special Conditions of Contract.
- (f) “Day” means calendar day of the Gregorian Calendar.
- (g) “Week” means a continuous period of seven (7) calendar days.
- (h) “Month” means calendar month of the Gregorian Calendar.
- (i) The “Contractor” means the individual or firm or company whether incorporated or not, undertaking the Works and shall include its legal representatives, or successors, or permitted assigns.
- (j) “Sub-Contractor” shall mean a person or a corporate body approved by Engineer-in-Charge who has entered into a contract with the Contractor to complete a part of the Works.

- (k) The “Contract Sum” / “Contract Price” means:
- (i) In the case of Lump Sum Contracts, the sum for which the bid is accepted.
 - (ii) In the case of item Rate Contracts, the cost of the Works arrived at after extension of the quantities shown in Schedule of Quantities by the item rates for the various items and summing them to arrive at the total price.
- (l) “Engineer-in-Charge” shall mean the person appointed by the Employer or his duly authorized representative who shall direct, supervise and be in charge of the Works for purposes of the Contract.
- (m) “Market rate” means the rate for an item of Work, determined on the basis of the cost of labour, and material brought to Site and incorporated in the Works and for use of plant and equipment for the Work executed plus 10% (ten percent) of the rate to cover all overheads and profits of the Contractor.
- (n) “Schedule(s) of Quantities” shall mean the priced and completed Schedule of Quantities forming the part of the bid or such Schedule of Quantities forming the part of the Contract, as the case may be, with amendments, if any, thereto.
- (o) The “Site” means the land and / or other places on, under, in or through which the Works is to be executed under the Contract including any other land or places which may be allotted by the Employer or used for the purposes of the Contract.
- (p) “Temporary/ Enabling Works” means all temporary works of every kind required in or about the execution, completion or maintenance of the Works.
- (q) “Urgent Works” shall mean any urgent measures which, in the opinion of the Engineer-in-Charge, become necessary during the progress of the Works to obviate any risk of accident or failure or which become necessary for reasons of security and safety.
- (r) “The Works” shall mean the Works to be executed in accordance with the Contract or part(s) thereof as the case may be and shall include all extra or additional, altered or substituted works or temporary/ enabling Works and urgent works as required for performance of the Contract.
- (s) “Cost” shall mean all expenditure reasonably incurred or to be incurred by the Contractor or Employer, as the case may be, whether on or off the Site, including overhead and other charges, but excluding profit.
- (t) Throughout these Bidding Documents, the term “Bid” and “Tender” and their derivatives (Bidder/Tenderer, Bidding/Tendering, Bidding Document/Tender Document, etc.); Bill of Quantity / Schedule of Quantity / Schedule of Items/Schedule of Quantities/ Bill of Quantities; Employer / NTPC ; Bid

Security / Earnest Money Deposit; Security Deposit / Performance Security/ Performance Guarantee; Engineer-in-Charge / Engineer, appearing anywhere in the Bidding Documents shall have the same meaning and are synonymous to each other.

- (u) Goods and Services Tax" or "GST" means taxes levied under the Central Goods and Services Tax Act, Integrated Goods and Services Tax Act, Goods and Services Tax (Compensation to States) Act and various State/Union Territory Goods and Services Tax Laws and applicable cesses, if any under the laws in force (hereinafter referred to as relevant GST Laws).

2. **Heading and Marginal Notes to Conditions**

Heading and marginal notes to these General Conditions of Contract shall not be deemed to form part thereof or be taken into consideration in the interpretation or construction thereof or of the Contract.

3. **Singular & Plural**

Where the context so requires, words importing the singular only will also include the plural and vice versa.

4. **Order of the precedence of the Documents**

- 4.1 The several documents forming the Contract are to be taken as mutually explanatory of one another, detailed drawings being followed in preference to small scale drawings, figured dimensions in preference to scale and Special Conditions in preference to General Conditions.
- 4.2 In the case of discrepancy between Schedule of Quantities, the Specifications and/or the Drawings, the following order of precedence shall govern:
 - (a) Description in Schedule of Quantities
 - (b) SCC
 - (c) Technical Specifications
 - (d) Drawings
- 4.3 If there are varying or conflicting provisions made within any document forming part of the Contract, the Engineer-in-Charge shall be the deciding authority with regard to the intention of the document.
- 4.4 Any error in description, quantity or rate in Schedule of Quantities or any omission there from shall not vitiate the Contract or release the Contractor from the execution of the whole or any part of the Works comprised therein according to drawings and specifications or from any of his obligations under the Contract.

5. **Instructions and Notices under the Contract:**

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5.1 Subject as otherwise provided in the Contract, all notices to be given on behalf of the Employer and all other actions to be taken on its behalf may be given or taken by the Engineer-in-Charge or any officer for the time being entrusted with the functions, duties and powers of the Engineer-in-Charge by the Employer. All instructions, notices and communications, etc., under the Contract shall be given in writing.

5.2 Contractor's Representative

5.2.1 The Contractor shall appoint its Representative within fourteen (14) days of the date of issuance of Letter of Award or before start of work whichever is earlier and shall request the Employer in writing to approve the person so appointed.

5.2.1.1 The Contractor's representative shall be a regular Employee/ Partner/ Director only and the Contractor shall be required to submit a Power of Attorney in original in favour of its representative. Notarized photocopy of the Power of Attorney shall be acceptable only if the Power of Attorney has been registered by the Contractor. The Employer may verify the photocopy of the Power of Attorney with the Original and the Contractor shall be required to produce the original Power of Attorney for verification, if required by the Employer. The relation of the Contractor's representative with the contractor such as Partner/ Employee/Director etc. should be clearly brought out in the Power of Attorney. The Contractor would be required to submit a documentary proof of the relation of the Contractor's representative with the contractor in the form of self attested copy of any of the following documents:

a) Previous financial year's Form 16 as available at TRACES site of Income tax department, if the POA holder is an employee of contractor. In case, the Contractor is not able to submit Form 16 (only in case of recent appointment or where Form 16 details are not uploaded at TRACES), the contractor would be required to submit an Appointment Letter/Salary Slip/other documentary evidence of the POA holder along with an affidavit stating the relationship between the POA holder and the Contractor along with. Further, the Contractor shall be required to submit the copy of Form 16 as available through TRACES site for every subsequent year also in respect of the POA holder till the period of authorization.

b) Articles of Association / Registered Partnership Deed if the POA holder is a partner or stake holder in Company.

5.2.1.2 In case, the Contractor's representative is also doing some other Contract(s)/Work(s) as nominee of the same contractor, the Contractor shall give a declaration citing list of all works where the Contractor's representative is the nominee.

5.2.1.3 If the Employer objects to the appointment giving the reason therefore, then the Contractor shall appoint a replacement within fourteen (14) days of such objection, and the foregoing provisions of this GCC Sub-Clause 5.2.1 shall apply thereto.

5.2.2 The Contractor's Representative shall represent and act for the Contractor at all times during the currency of the Contract and shall communicate and provide to the Engineer-in-charge all the Contractor's notices, instructions, information and all other communications under the Contract.

All notices, instructions, information and all other communications given by the Employer or the Engineer-in-charge to the Contractor under the Contract shall be given to the Contractor's Representative or, in its absence, its deputy, except as herein otherwise provided. The Contractor or its authorised representative shall be in attendance at the Site(s) during all working hours and

shall superintend the execution of the Works with such additional assistance in each trade as the Engineer-in-Charge may consider necessary. Orders given to the Contractor's representative shall be considered to have the same force as if they had been given to the Contractor.

The Contractor shall not revoke the appointment of the Contractor's Representative without the Employer's prior written consent, which shall not be unreasonably withheld. If the Employer consents thereto, the Contractor shall appoint some other person as the Contractor's Representative, pursuant to the procedure set out in GCC Sub-Clause 5.2.1.

- 5.2.3 The Contractor's Representative may, subject to the approval of the Employer (which shall not be unreasonably withheld), at any time delegate to any person any of the powers, functions and authorities vested in him or her. Any such delegation may be revoked at any time. Any such delegation or revocation shall be subject to a prior notice signed by the Contractor's Representative, and shall specify the powers, functions and authorities thereby delegated or revoked. No such delegation or revocation shall take effect unless and until a copy thereof has been delivered to the Employer and the Engineer-in-charge.

Any act or exercise by any person of powers, functions and authorities so delegated to him or her in accordance with this GCC Sub-Clause 5.2.3 shall be deemed to be an act or exercise by the Contractor's Representative.

- 5.2.4 The Employer may by notice to the Contractor object to any representative or person employed by the Contractor in the execution of the Contract who, in the reasonable opinion of the Employer, may behave inappropriately, may be incompetent or negligent, or may commit a serious breach of the Safety regulations provided under GCC Sub-Clause 10.4.9. The Employer shall provide evidence of the same, whereupon the Contractor shall remove such person from the Facilities.

- 5.2.5 If any representative or person employed by the Contractor is removed in accordance with GCC Sub-Clause 5.2.4, the Contractor shall, where required, appoint a replacement as per clause 5.2.1.

- 5.2.6 In case any of the information/declaration/undertaking provided by Contractor/Contractor's representative is found to be false and/or the contractor/Contractor's representative suppresses any relevant information at any stage, the Contractor will be liable for actions in terms of Employer's Banning policy.

- 5.3 The Engineer-in-Charge shall communicate or confirm the instructions to the Contractor in respect of the execution of Work in a 'Works Site Order Book' maintained in the office of the Engineer-in-Charge and the Contractor or his authorised representative shall confirm receipt of such instructions by signing the relevant entries in this Book. If required by the Contractor, he shall be furnished a certified true copy of such instruction(s).

6. **Laws governing the Contract**

- 6.1 This Contract shall be governed by the Indian laws for the time being in force. The Courts at New Delhi shall have jurisdiction, in all matters unless otherwise stated in the SCC.

7. **Settlement of Disputes**

7.1 **Mutual Consultation**

If any dispute of any kind whatsoever shall arise between the Employer and the Contractor in connection with or arising out of the Contract, including without prejudice to the generality of the foregoing, any question regarding its existence, validity or termination, or the execution of the

Works, whether during the progress of the Works or after their completion and whether before or after the termination, abandonment or breach of the Contract, the parties shall seek to resolve any such dispute or difference by mutual consultation.

If the parties fail to resolve such a dispute or difference by mutual consultation, then the dispute may be settled through Expert Settlement Council / Arbitration / other remedies available under the applicable laws.

7.2. Resolution of Dispute through Expert Settlement Council

If the parties fail to resolve such a dispute or difference by mutual consultation, the dispute if the parties agree, may be referred to Conciliation in cases involving disputed amount up to Rs 250 crores, which is to be arrived at considering the claim and counter claim of the parties to the dispute.

7.2.1 Invitation for Conciliation:

7.2.1.1 A party shall notify the other party in writing about such a dispute it wishes to refer for Conciliation within a period of 30 days from the date of raising of the dispute in case of failure to resolve the same through mutual consultation. Such Invitation for Conciliation shall contain sufficient information as to the dispute to enable the other party to be fully informed as to the nature of the dispute, amount of the monetary claim, if any, and apparent cause of action.

7.2.1.2 Upon acceptance of the invitation to conciliate, the other party shall submit its counter claim, if any, within a period of 30 days from the date of the invitation to conciliate. If the other party rejects the invitation or disputed amount exceeds Rs 250 crores, there will be no Conciliation proceedings. There shall be no Conciliation where claim amount is only up to Rs 5 lakhs.

7.2.1.3 If the party initiating Conciliation does not receive a reply within thirty days from the date on which it sends the invitation, or within such other period of time as specified in the invitation, it shall treat this as a rejection of the invitation to conciliate from the other party.

7.2.2 Conciliation:

7.2.2.1 Where Invitation for Conciliation has been furnished under GCC sub clause 7.2.1, the parties shall attempt to settle such dispute through Expert Settlement Council (ESC) which shall be constituted by CMD, NTPC.

7.2.2.2 ESC will be formed from experts comprising of three members from the panel of conciliators maintained by NTPC. However, there will be single member ESC for disputes involving claim and counter claim (if any) up to Rs. 1 crore. CMD will have authority to reconstitute an ESC to fill any vacancy.

7.2.2.3 The eligible persons for consideration for empanelment in the panel of conciliators shall be amongst Retired Civil Servants of Govt. of India not below the rank of Joint Secretary, Retired Judges, Retired Executive directors/Directors/ Chairman of any Maharatna / Navratna company in India other than NTPC Ltd, Retired Independent Directors who have served on the Board of any Maharatna / Navratna company in India other than NTPC Ltd and Independent experts in their respective fields preferably registered with the Indian Council of Arbitration or Delhi International Arbitration Centre or Federation of Indian Chambers of Commerce and Industry or SCOPE Arbitration Forum.

7.2.3 Proceedings before ESC:

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7.2.3.1 The claimant shall submit its statement of claims along with relevant documents to ESC members, and to the party(s) indicated in the appointment letter within 30 days of the issue of the appointment letter. The respondent shall file its reply and counter claim (if any) within 30 days of the receipt of the statement of claims. Parties may file their rejoinder/additional documents, if any in support of their claim/counterclaim within next 15 days. No documents shall be allowed thereafter, except with the permission of ESC.

7.2.3.2 The parties shall file their claim and counterclaim in the following format

- a. Chronology of the dispute
- b. Brief of the contract
- c. Brief history of the dispute
- d. Issues

Sl. No.	Description of Claims/ Counter claims	Amount (in foreign currency/INR)	Relevant Contract Clause

- e. Details of Claim(s)/Counter Claim(s)
- f. Basis/Ground of claim(s)/counter claim(s) (along with relevant clause of contract)

Note: Statement of claims shall be restricted to maximum limit of 20 pages.

7.2.3.3 In case of 3 members ESC, 2 members will constitute a valid quorum and the meeting can take place to proceed in the matter after seeking consent from the member who is not available. However, ESC recommendations will be signed by all the members.

7.2.3.4 The parties shall be represented by their in house employees. No party shall be allowed to bring any advocate or outside consultant/advisor/agent to contest on their behalf. Ex-officers of NTPC who have handled the subject matter in any capacity shall not be allowed to attend and present the case before ESC on behalf of contractor. However, ex-employees of parties may represent their respective organizations. Parties shall not claim any interest on claims/counter-claims from the date of notice invoking Conciliation till execution of settlement agreement, if so arrived. In case, parties are unable to reach a settlement, no interest shall be claimed by either party for the period from the date of notice invoking Conciliation till the date of ESC recommendations and 30 days thereafter in any further proceeding.

7.2.3.5 ESC will conclude its proceedings in maximum 10 meetings, and give its recommendations within 90 days of its first meeting. ESC will give its recommendations to both the parties recommending possible terms of settlement. CMD, NTPC may extend the time/number of meetings, in exceptional cases, if ESC requests for the same with sufficient reasons.

7.2.3.6 Depending upon the location of ESC members and the parties, the venue of the ESC meeting shall be either Delhi/Mumbai/Kolkata/Chennai or any other city whichever is most economical from the point of view of travel and stay etc. All the expenditure incurred in ESC proceedings shall be shared by the parties in equal proportion.

7.2.4 Fees & Facilities to the Members of the ESC

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The cost of Conciliation proceedings including but not limited to fees for Conciliator, Airfare, Local transport, Accommodation, cost towards conference facility etc shall be as provided herein below:

S. No.	Fees/ Facility	Entitlement
1	Fees	As paid to NTPC Independent Directors [Presently Rs. 20,000 per meeting]. In addition, each conciliator to be paid Rs. 10,000 for attending meeting to authenticate the settlement agreement - max. of Rs. 2,10,000 per case per Conciliator.
2	Secretarial expenses	Rs. 10,000 lump sum (to 1 member only).
3	Transportation in the city of the meeting	Car as per entitlement or Rs. 2,000 per day
4	Venue for meeting	NTPC conference rooms
Facilities to be provided to the out-stationed member		
5	Travel from the city of residence to the city of meeting	As per entitlement of Independent Directors. Executive class air tickets / first class AC train tickets/ Luxury car/ reimbursement of actual fare. However, entitlement of air travel by Business class shall be subject to austerity measures, if any, ordered by Govt. of India.
6	Transport to and fro airport/ railway station in the city of residence	Car as per entitlement or Rs. 3,000
7	Stay for out stationed members	As per entitlement of Independent Directors.
8	Transport in the city of meeting	Car as per entitlement or Rs. 2000 per day

Aforesaid fees is subject to revision by NTPC from time to time and subject to government guidelines on austerity measures, if any. All the expenditure incurred in the ESC proceedings shall be shared by the parties in equal proportions. The Parties shall maintain the account of expenditure and present to the other for the purpose of sharing on conclusion of the ESC proceedings.

7.2.5 If decision of NTPC is acceptable to the contractor, a Settlement Agreement under section 73 of the Arbitration and Conciliation Act 1996 will be signed within 15 days of contractor's acceptance and same shall be authenticated by all the ESC members. Parties are free to terminate Conciliation proceedings at any stage as provided under the Arbitration and Conciliation Act 1996.

7.2.6 The parties shall keep confidential all matters relating to the Conciliation proceedings. Parties

shall not rely upon them as evidence in arbitration proceedings or court proceedings.

7.3 **Arbitration**

7.3.1 If the process of mutual consultation and/or ESC fails to arrive at a settlement between the parties as mentioned at GCC Sub-Clauses 7.1 & 7.2 above, Employer or the Contractor may, within Thirty (30) days of such failure, give notice to the other party, with a copy for information to the ESC (as applicable), of its intention to commence arbitration, as hereinafter provided, as to the matter in dispute, and no arbitration in respect of this matter may be commenced unless such notice is given. The mechanism of settling the disputes through arbitration shall be applicable only in cases where the disputed amount (i.e. total amount of Claims excluding claims of interest) does not exceed Rs. 25 crores. In case the disputed amount exceeds Rs. 25 Crores, the parties shall be within their rights to take recourse to remedies as may be available to them under the applicable laws other than Arbitration after prior intimation to the other party. There shall be no arbitration where the claim amount is only up to Rs. 5 lakhs.

The parties at the time of invocation of arbitration shall submit all the details of the claims and the counter-claims including the Heads/Sub-heads of the Claims/Counter-Claims and the documents relied upon by the parties for their respective claims and counter-claims. The parties shall not file any documents/details of the claims and counter-claims thereafter.

The claims and the counter claims raised by the parties at the time of invocation of the arbitration shall be final and binding on the parties and no further change shall be allowed in the same at any stage during arbitration under any circumstances whatsoever.

The parties to the contract shall invoke arbitration within Six months from the date of completion of the execution of work under the contract or the termination of the contract as the case may be and the parties shall not invoke arbitration later on after expiry of the said period of six months. The parties shall not invoke arbitration other than in the case of completion of execution of work or the termination of the contract as mentioned above.

Notwithstanding the above, in case of disputes with Indian Contractor who is a Central Government Department /Enterprise /organisation or a State Level Public Enterprise (SLPE), the aforesaid limit of Rs 25 crores shall not be applicable and arbitration proceeding may be commenced irrespective of the amount involved in dispute if the dispute could not be resolved through Conciliation as brought out at GCC Sub Clause 7.2 above.

7.3.2 Any dispute in respect of which a notice of intention to commence arbitration has been given, in accordance with GCC Sub Clause 7.3.1, shall be finally settled by arbitration.

7.3.3 Any dispute raised by a party to arbitration shall be adjudicated by a Sole Arbitrator appointed by CMD, NTPC from the List of empanelled Arbitrators of NTPC in the following manner:

- (a) A party willing to commence arbitration proceeding shall invoke Arbitration Clause by giving 60 days notice to the other party.
- (b) If the Arbitrator so appointed dies, resigns, becomes incapacitated or withdraws for any reason from the proceedings, it shall be lawful for CMD, NTPC to appoint another person in his place in the same manner as aforesaid. Such person shall proceed with the reference from the stage where his predecessor had left.
- (c) It is agreed between the parties that the Arbitration proceedings shall be conducted as

per the provisions of Fast Track Procedure as provided under section 29B of the Arbitration and Conciliation Act, 1996 as amended.

- (d) Arbitrator shall be paid fees at the following rates:

Amount of Claims and Counter Claims (excluding interest)	Lump sum fees (including fees for study of pleadings, case material, writing of the award, secretarial charges etc.) to be shared equally by the parties.
Upto Rs 50 lakhs	Rs. 10,000 per meeting subject to a ceiling of Rs. 1,00,000/-.
Above Rs 50 lakhs to Rs 1 crore	Rs. 1,35,000/- plus Rs. 1,800/- per lakh or a part there of subject to a ceiling of Rs. 2,25,000/-.
Above Rs. 1 crore and upto Rs. 5 Crores	Rs. 2,25,000/- plus Rs. 33,750 per crore or a part there of subject to a ceiling of Rs. 3,60,000/-.
Above Rs. 5 crores and upto Rs. 10 crores.	Rs. 3,60,000/- plus Rs. 22,500/- per crore or a part there of subject to a ceiling of Rs. 4,72,500/-.
Above Rs. 10 crores	Rs. 4,72,500 plus Rs. 18,000/- per crore or part thereof subject to a ceiling of Rs. 10,00,000/-.

If the claim is in foreign currency, the SBI Bills Selling Exchange rate prevailing on the date of claim shall be used for the purpose of converting the claim in Indian Rupee which may be used for determining the arbitration fee as brought out above.

- e) If after commencement of the Arbitration proceedings, the parties agree to settle the dispute mutually or refer the dispute to Conciliation, the arbitrator shall put the proceedings in abeyance until such period as requested by the parties. Where the proceedings are put in abeyance or terminated on account of mutual settlement of dispute by the parties, the fees payable to the arbitrator shall be determined as under:
- (i) 40% of the fees if the Pleadings are complete.
 - (ii) 60% of the fees if the Hearing has commenced.
 - (iii) 80% of the fees if the Hearing is concluded but the Award is yet to be passed.
- f) Each party shall pay its share of arbitrator's fees in stages as under:
- (i) 40 % of the fees on Completion of Pleadings.
 - (ii) 40% of the fees on Conclusion of the Final Hearing.
 - (iii) 20% at the time when arbitrator notifies the date of final award.
- g) The Claimant shall be responsible for making all necessary arrangements for the travel/ stay of the Arbitrator including venue of arbitration, hearings. The parties shall share the expenses for the same equally.
- h) The Arbitration shall be held at Delhi only.
- i) The Arbitrator shall give reasoned and speaking award and it shall be final and binding on the parties.

- j) Subject to the aforesaid conditions, provisions of the Arbitration and Conciliation Act, 1996 and any statutory modifications or re-enactment thereof shall apply to the arbitration proceedings under this clause.

7.3.4 In case the Indian Contractor is a Central Government Department/Enterprise/organisation or a State Level Public Enterprise (SLPE), the dispute arising between the Employer and the Contractor shall be referred for resolution to the Permanent Machinery of Arbitrators (PMA) of the Department of Public Enterprises, Government of India as per Office Memorandum No. 4(1) 2011-DPE(PMA)-GL dated 12.06.2013 issued by Government of India, Ministry of Heavy Industries and Public Enterprises, Department of Public Enterprises and its further modifications and amendments.

7.4 Notwithstanding any reference to the Conciliation or Arbitration herein,

- (a) the parties shall continue to perform their respective obligations under the Contract unless they otherwise agree.
- (b) the Employer shall pay the Contractor any monies due to the Contractor.

B. Subject Matter of Contract

8. Scope of Contract

The Work to be carried out under the Contract shall be as delineated in Bidding Documents and shall, except as otherwise provided in these conditions, include all labour, materials, tools, plant, equipment, and transport which may be required in preparation of and for and in the full and entire execution and completion of the Works.

9. Notices to Local Bodies

9.1 The Contractor shall comply with and give all notices required under any Governmental authority, instrument, rule or order made under any Act of Parliament, State Laws or any regulation or bye-laws of any local authority relating to the Works. He shall before making any variation from the Contract drawings necessitated by such compliance give to the Engineer-in-Charge a written notice giving reasons for the proposed variation and obtain the Engineer-in-Charge's instructions thereon, in writing.

9.2 The Contractor shall pay and indemnify the Employer against any liability in respect of any fees or charges payable under any Act of Parliament, State laws or any Government instrument, rule or order and any regulations or bye-laws of any local authority in respect of the Works.

10. Human Resources

10.1 The Contractor for the purpose of the Contract shall engage / employ adequate number of key personnel in all areas such as design / engineering (wherever applicable), planning, scheduling and construction and carrying out of all maintenance of his plant and equipment (as detailed in the SCC) and competent

and skilled work force as directed by the Engineer-in-Charge. The Engineer-in-Charge will approve any proposed replacement of such key personnel including work force only if their qualifications, experience, competence and capabilities are substantially equal to or better than those personnel originally identified and approved by the Engineer-in-Charge.

10.2 The Engineer-in-Charge may require the Contractor to remove from Site of Works or from any other area of Work related to the Contract, any member of the Contractor personnel or work force who

- (i) Persists in any misconduct or lack of care
- (ii) Performs his duties incompetently or negligently or otherwise carelessly
- (iii) Fails to conform with any provisions of the Contract or
- (iv) Persists in any conduct which is prejudicial to the safety, health or protection of the Work and environment.

If appropriate, the Contractor shall appoint a suitable replacement within fourteen (14) days or within such period as may be agreed between the Engineer-in-Charge and Contractor.

10.3 The Contractor shall unless otherwise provided in the Contract, make his own arrangement for engagement of all staff and labour, local or otherwise and for their payment, housing, transport, lodging and welfare as may be required by law and or by industry practice. The Contractor shall provide the Engineer-in-Charge a return in detail in such form and at such intervals as he may reasonably prescribe showing the staff and number of the several classes of labour and other staff from time to time employed by the Contractor at Site or in connection with the Work along with such information as the Engineer-in-Charge may reasonably require.

10.4 **Labour laws and Regulations and compliance thereof**

10.4.1 During the entire period of Contract, the Contractor and his Sub-Contractors shall, at all times abide by all existing labour enactments, rules made therein, regulations, notifications and bye-laws by the appropriate government, local authority or any other labour laws or notification that may be issued under any labour law prevailing as on the date seven(7) days prior to the deadline set for submission of the bids, published by the State or Central Government or Local Authorities. An illustrative list of applicable acts, notifications, rules etc. in connection with the labour as applicable is provided in SCC. This list is not in any way exhaustive and shall not absolve the Contractor from any of his liabilities or responsibilities in compliance with any other laws, regulations, notifications that may be in force during the tenure of Contract.

10.4.1.1 During the entire period of Contract, the Contractor and his Sub-Contractors shall, at all times abide by the following Acts/Statutes related to Human Resources:

1. Factories Act, 1948; Contract Labour (Regulation & Abolition) Act, 1970;
2. EPF & MP Act, 1952;

3. Building & Other Construction Workers (Regulation of Employment & Conditions of Service) Act, 1996;
4. ESI Act, 1948;
5. Minimum Wages Act, 1948;
6. Payment of Wages Act, 1936
7. Payment of Bonus Act, 1965;
8. Payment of Gratuity Act, 1972;
9. Workmen's Compensation Act, 1923;
10. ID Act, 1947;
11. Maternity Benefit Act, 1961;
12. Inter-State Migrant Workmen (Regulation of Employment & Conditions of Service) Act, 1979;
13. Fatal Accidents Act, 1855
14. Model Welfare Code

The above will deem to include all relevant/applicable rules made thereunder, regulations, notifications and bye laws of the State or Central Govt. or the local authority and any other labour law (including rules) regulations, bye laws as well as those that may be passed or notification that may be issued under any labour law present and in future either by State or Central Govt. or by local authority.

- 10.4.2 The Contractor and his Sub-Contractors shall indemnify the Employer, from any action taken against the Employer by any competent authority in connection with the enforcement of the applicable laws, regulations, notifications, on account of contravention of any of the provisions therein, including amendments thereto. If the Employer is caused to pay or otherwise made liable, such amounts as may be necessary for non-observance of the provisions stipulated in the laws, rules, notifications including amendments, if any on the part of the Contractor and/or his Sub-Contractors, the Engineer-in-Charge / Employer shall have the right to deduct any such money from any amount due to the Contractor including his performance security, under the Contract. The Employer shall also have the right to recover from the Contractor any sum required or estimated as required for making good any loss or damage suffered / likely to be suffered by the Employer, on this account.
- 10.4.3 If due to an enactment of any new Act or Statute and rules made thereunder or any modification to the Acts/Statute or rules made thereunder, all after seven (7) days prior to the deadline set for submission of the bids and as a consequence thereof, the Contractor has to incur additional cost or expenditure, the same will be reimbursed by the Employer to the Contractor, excepting those due to reasons attributable to the Contractor and those being already compensated by other provisions of the Contract, like Price Adjustment, Taxes and Duties etc.
- 10.4.4 It is specifically agreed that the Contractor and his Sub-Contractors shall obtain all the necessary registration, licenses, permits, authorisations etc. required under various enactments / Regulations enforced from time to time, specifically registration as employer under Provident Fund Act and Contract Labour Regulation & Abolition Act, and the Employer shall not be liable for any violation by the Contractor in this regard.
- 10.4.5 The employees of the Contractor or his Sub-Contractor(s) shall in no case be treated as the employees of the Employer at any point of time.
- 10.4.6 The Contractor and his Sub-Contractors shall be liable to make all due payments to

all their employees and ensure compliance with labour laws. If the Employer, is held liable as 'PRINCIPAL EMPLOYER' or otherwise to incur any expenditure or to make any contributions under any legislation of the Government or Court decision, in respect of the employees of the Contractor or his Sub-Contractors, then the Contractor would reimburse the amounts of such expenditure/contribution so made by the Employer.

10.4.7 In case the ESI act is not applicable to the area where the Work is executed, as evidenced by the Certificate/Letter submitted to this effect from the local authorities, the Contractor shall be liable to arrange and pay for the expenses towards the medical treatment in respect of all labour employed by him for the execution of the Contract.

10.4.8 Staff quarters & labour hutments at Site if available, may be provided to the Contractor on chargeable basis at the discretion of the Engineer-in-Charge on mutually agreed terms and conditions.

10.4.9 **Safety**

The Contractor, including his sub-contractors, while executing the Works, will strictly comply with the statutory requirements (including amendments thereof), as applicable, in respect of safety of his employees, equipment and materials. The contractor will also comply with the provisions of NTPC Safety Rules as issued from time to time and displayed on NTPC's tender website <http://www.ntpctender.com>. Within 30 days from the date of Letter of award, the Contractor, shall develop and submit for the Employer's approval, his own Safety, Health and Environment policy in line with the above-mentioned statutory requirements and NTPC rules. The detailed requirements to be complied by the Contractor with regard to the safety of his personnel, equipment and materials are enumerated in the Technical Specifications.

11. **Shift Work**

11.1 To achieve the required rate of progress in order to complete the Works within the Time for Completion, the Contractor may carry on the work, round the clock, in multiple shifts per day, as may be necessary. The Contractor shall however be responsible to comply with all applicable laws in this regard.

11.2 No additional payment will be made on account of round the clock working in multiple shifts.

11.3 Wherever the work is carried out at night adequate lighting of working areas and access routes for pedestrians or vehicles shall be provided by the Contractor at his cost. Sufficient notice should be given by the Contractor to the Engineer-in-Charge regarding the details of works in shifts so that necessary supervision could be provided.

12. **Cooperation with other Contractors/Agencies**

12.1 The Contractor shall extend all reasonable cooperation to other Contractors, agencies etc. of the Employer engaged in connection with the Work or any other

Work not in the scope of this Work as may be required by the Engineer-in-Charge.

- 12.2 The Contractor shall attend at his cost, all the meetings with the Engineer-in-Charge, other contractors and the Consultants of the Employer for the purposes of the Contract. The Contractor shall attend such meetings as and when required by the Engineer-in-Charge.

13. **Materials obtained from Excavation**

- 13.1 Materials of any kind obtained from excavation on the Site shall remain the property of the Employer and shall be disposed of as the Engineer-in-Charge may direct.

14. **Treasure, Trove, Fossils, etc**

- 14.1 All fossils, coins, articles of value or antiquity and structures and other remains or things of geological or archaeological interest discovered on the Site shall be the absolute property of the Employer and the Contractor shall take reasonable precautions to prevent his workmen or any other person from removing or damaging any such article or thing, shall immediately upon discovery thereof and before removal acquaint the Engineer-in-Charge with such discovery and carry out the Engineer-in-Charge's directions as to the disposition of the same, at the cost of the Employer.

15. **Protection of Trees**

- 15.1 Trees designated by the Engineer-in-Charge shall be protected from damage during the course of the Works and earth level within at least one (1) meter of each such tree shall not be disturbed. Where necessary, such trees shall be protected by providing temporary fencing at the cost of the Employer.

16. **Security Watch and Lighting**

- 16.1 The Contractor shall provide and maintain at his own expense all lights, guards, fencing and watching when and where necessary or required by the Engineer-in-Charge for the protection of the Works or for the safety and convenience of those employed on the Works or the public.

17. **Prevention of Pollution**

- 17.1 The Contractor shall make necessary arrangement to prevent pollution of the water in any adjacent water bodies including stream, springs, nallah, river and lakes etc. The Contractor shall be solely responsible and liable for all damage caused by any pollution that may take place during the execution of the Work.

18. **Explosives**

- 18.1 Permission for the use of explosives shall be obtained from the Engineer-in-Charge or from any appropriate authority as directed by the Engineer-in-Charge and all

explosive materials shall be used only under close supervision. It shall be the responsibility of the Contractor to seek and obtain any necessary permits, and to ensure that the requirements of the authorities are complied with, in all respects. Failure to do so may result in the Engineer-in-Charge withdrawing permission to use explosives. The indemnification provided for, under the General Condition of Contract shall include indemnification against all claims in respect of any incident arising from the use of explosives.

19. Training of Apprentices:

19.1 The Contractor shall during the currency of the Contract when called upon by the Engineer-in-Charge engage and also ensure engagement by Sub-Contractors employed by the Contractor in connection with the Works, such number of Apprentices in the categories and for such periods as may be reasonably required by the Engineer-in-Charge. The Contractor shall train them as required under the Apprentices Act, 1961 and shall be responsible for all obligations of the Employer under the Act including the liability to make payment to apprentices as required under the Act.

20. Handing over of Site

20.1 The Employer shall make available the Site to the Contractor as soon as possible after the award of the Contract free of encumbrance. The Contractor shall not be permitted to enter on (other than for inspection purposes) or take possession of the Site until instructed to do so by the Engineer-in-Charge in writing.

20.2 The Employer reserves the right to hand over the Site in parts progressively to the Contractor. The Contractor will be required to take possession of the Site without any undue delay and do Work on the released fronts in parts without any reservation whatsoever.

20.3 However, in case of any delay in handing over of the Site to the Contractor, which delays the performance of the Work, commensurate to the resources mobilised by the Contractor, then the Contractor will be eligible for suitable extension in time for completion of the Works or any other compensation as per the provisions of GCC Sub-Clause 29.7.

20.4 The portion of the Site to be occupied by the Contractor shall be defined and/or marked on the Site plan, failing which these shall be indicated by the Engineer-in-Charge at Site and the Contractor shall on no account be allowed to extend his operations beyond these areas. Further, the Contractor shall not hinder in any way the working of other contractors on the Site.

20.5 The Contractor shall plan his work as per available roads at Site. Wherever, the access roads near to the Work Site are not available, the Contractor shall make his own arrangement for approach to the Work Site including borrow/disposal areas and for movement of men, machinery, other equipment etc. required for carrying out the Works included under this Contract. All drainage of Works area and all weather truckable/ haulage roads as required by the Contractor shall be

constructed and maintained during the construction period by the Contractor at his own cost.

20.6 The Contractor shall provide, if necessary, all temporary access to the Work Site and shall alter, adapt and maintain the same as required from time to time and shall take up and clear them away as and when no longer required and, as and when ordered by the Engineer-in-Charge and make good all damage done to the Site.

20.7 The Contractor shall be permitted the usage of all access roads built by the Employer for the purposes of the Contract. If any damage is done to such roads and associated areas like drains, culverts and the like, by the Contractor, the same shall be made good by the Contractor at his own cost, but as may be directed by the Engineer-in-Charge.

21. Duties and Powers of Engineer-in-Charge and his representatives

21.1 The Engineer-in-Charge shall have the full powers in respect of all the matters in connection with or arising out of this Contract, excepting those specifically reserved for the Employer. However, the Engineer-in-Charge shall not have any power to relieve the Contractor of any of his obligations and responsibilities under the Contract.

21.2 The duties of the representative of the Engineer-in-Charge are to watch and supervise the Works and to test and examine any materials to be used or workmanship employed in connection with the Works. He shall have no authority to order any work involving any extra payment by the Employer nor to make any variation in the Works, creating a financial liability to the Employer.

21.3 The Engineer-in-Charge may from time to time in writing delegate to his representative any of the powers and authorities vested in the Engineer-in-Charge and shall furnish to the Contractor a copy of all such written delegation of powers and authorities. Any written instruction or written approval given by the representative of the Engineer-in-Charge to the Contractor within the terms of such delegation shall bind the Contractor and the Engineer-in-Charge as though it has been given by the Engineer-in-Charge.

21.4 Failure of the Representative of the Engineer-in Charge to disapprove any work or materials shall not prejudice the power of the Engineer-in-Charge thereafter to disapprove such Work or materials and to order the pulling down, removal or breaking up thereof.

21.5 If the Contractor shall be dissatisfied with any decision of the representative of the Engineer-in-Charge he shall be entitled to refer the matter to the Engineer-in-Charge who shall there-upon confirm, reverse or vary such decision.

C. Payment

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22. Contract Performance Security

22.1.1 The Contractor shall within twenty-one (21) days of the Letter of Award, provide a Contract Performance Guarantee (CPG) towards faithful performance of the Contract for ten percent (10%) of the Contract Price. Alternatively, the Contractor may furnish the Contract Performance Guarantee in two Stages viz. A Contract Performance Guarantee for an amount equivalent to 5% of the Contract Price within twenty-one (21) days from the date of Letter of Award to be followed up with another Contract Performance Guarantee for an amount equivalent to 5% of the Contract Price prior to the scheduled completion of 50% of total scope of work.

22.1.2 The Contractor also has the option to furnish an Initial Contract Performance Guarantee (ICPG) equivalent to a minimum of 2% of the Contract Price. In case of such a Contractor, who furnishes Initial Contract Performance Guarantee, the Employer shall at the time of making any payment to him for the work done under the Contract deduct towards the Contract Performance Security an amount equivalent to 10% of the gross bill amount accepted for payment until the Contract Performance Security so deducted including the amount of Initial Contract Performance Security becomes equal to 10% of the Contract Price, subject to the condition that whenever the Contract Performance Security so deducted reaches a limit of 2% of the Contract Price, the Contractor may convert the amount into a Bank Guarantee as aforesaid.

22.1.3 The Contract Performance Guarantee/Initial Contract Performance Guarantee shall be in the form of a Bank Guarantee issued by a Bank from the list enclosed in the Bidding Documents. The format of the said bank guarantee shall be in accordance with the format included in the Section VIII (Forms and Procedures). This bank guarantee shall have an initial validity upto ninety (90) days beyond the completion of Defect Liability Period of the Contract.

However, in case the date of completion of defects liability period gets extended, the validity of the Contract Performance Guarantee shall be extended by the period of extension of completion of defects liability.

22.2 One half (50%) of the Contract Performance Security amount shall be refunded/released to the Contractor on the issuance of Completion Certificate by the Engineer-in-Charge certifying the completion of entire Works under the Contract. The Engineer-in-Charge shall on demand from the Contractor, release/refund the remaining portion of the Contract Performance Guarantee on expiry of the Defects Liability Period or on payment of the amount of the Final Bill payable, whichever is later, provided the Engineer-in-Charge is satisfied that there is no demand outstanding against the Contractor, arising out of the Contract.

22.3 No interest shall be payable to the Contractor against the Contract Performance Guarantee furnished/recovered by way of deductions from running account payments from the Contractor, by the Employer.

23. Advance

23.1 Mobilisation Advance

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- i) A lumpsum mobilisation advance not exceeding 4% of the Contract Value as awarded shall be paid to the Contractor at the rate of interest mentioned in SCC subject to the following conditions:
- a) On unconditional acceptance of Letter of Award by the Contractor.
 - b) Acceptance and finalisation of detailed Work Schedule and finalisation of equipment mobilization schedule.
 - c) Submission of an irrevocable bank guarantee for the amount equivalent to 110% of the advance plus GST as applicable on the advance payment to be paid to the contractor in the proforma of bank guarantee provided in Section-VIII (Forms and Procedures) - Proforma of Bank Guarantee for Lumpsum Advance from any bank from the list enclosed in Bidding Documents and acceptable to the Employer.
 - d) Finalisation of Quality Assurance Programme in accordance with GCC Clause entitled 'Quality Assurance programme'.
 - e) Submission of an unconditional, irrevocable bank guarantee for the amount of Initial Contract Performance Security Deposit/ Contract Performance Security Deposit from any bank from the list enclosed in Bidding Documents and acceptable to the Employer in the proforma of bank guarantee provided in Section VIII (Forms and Procedures)- Form of Bank Guarantee for Contract Performance Guarantee.
 - f) On completion of the requirement of Qualifying Requirements, the documentation in respect of associate(s), if applicable, in line with the provisions thereof in BDS, and on submission of an unconditional and irrevocable bank guarantee from any bank as per list enclosed in Bidding Documents and acceptable to Employer in the proforma provided in Section VIII (Forms and Procedures).
 - g) The advance payment shall be released only after the verification and confirmation of the above bank guarantees from the issuing bank.
- ii) A further lumpsum mobilisation advance not exceeding 4% of the Contract Value as awarded shall be paid to the Contractor from Project Site at the rate of interest mentioned in SCC subject to the following conditions:
- a) Fulfilment of conditions mentioned at GCC Sub-Clause 23.1 (i) a, b, d, e & f.
 - b) On certification of the Engineer-in-Charge that the Contractor has opened his office at Site and mobilised the plant and equipment specifically identified in Letter of Award of the Contract for releasing this advance.

- c) Submission of an irrevocable bank guarantee for the amount equivalent to 110% of the advance plus GST as applicable on the advance payment to be paid to the contractor in the proforma provided in Section-VIII (Forms and Procedures) - Proforma of Bank Guarantee for Lumpsum Advance from any bank from the list enclosed in Bidding Documents and acceptable to the Employer.
- d) The advance payment shall be released only after the verification and confirmation of the above bank guarantees from the issuing bank.
- e) Mobilisation advance shall be released after certification of Engineer-in-Charge that the Contractor has brought to site the Safety equipment & Safety Personal Protective Equipment as per minimum quantity specified in the Bidding Documents.

In case the Contractor decides not to take Mobilisation advance, the first progressive payment shall be released after certification of Engineer-in-Charge that the Contractor has brought to site the Safety equipment & safety Personal Protective Equipment as per minimum quantity specified in the Bidding Documents.

23.2 **Equipment Advance**

An amount not exceeding 75% of the purchase price of the new plant and equipment specifically acquired for the Work and brought to Site against production of documents in support thereof and subject to the condition that the Engineer-in-Charge considers the price reasonable and that such plant and equipment are necessary for the Works and not in excess of requirements and are hypothecated in favour of the Employer in the form required by the Employer shall be payable. "In case the Contractor is an entity other than a Sole Proprietorship Concern or a Partnership Firm, the Contractor shall get the Hypothecation Deed thus executed, registered with the Registrar of Companies in the manner and timeframe required as per the Companies Act, 2013 of India. "Interest on the advance outstanding, shall be levied at the percentage mentioned in SCC. Total advance amount against plant and equipment shall in no case exceed 6% of the total contract value as awarded. Provided that if a request for loan is made by the Contractor against both the aforementioned provisions of this condition, viz. GCC Sub-Clauses 23.1 and 23.2, the total sum to be given as loan shall not exceed 10% of the total contract value as awarded.

- 23.3 All the above advances mentioned at GCC Sub-Clause 23.1(i), 23.1 (ii) and 23.2 shall be interest bearing and the rate of interest for mobilization advance and equipment advance shall be as specified in SCC.
- 23.4 Any of the above advances may be claimed by the Contractor at his option in instalments.
- 23.5 The above advance shall be utilised by the Contractor for the purposes of this Contract only and for no other purpose.

- 23.6 All payments of advances may be released electronically, wherever technically feasible.
- 23.7 The recovery of the advances as above and interest thereon shall be made by deduction from the on account payments referred to in GCC Clause entitled 'Payment on Account/Progressive Interim Payments' in suitable percentages fixed by the Engineer-in-Charge in relation to the total cumulative advances released and the progress of the work, so that all the sums advanced with interest thereon shall be fully recovered starting from the time 10% and ending by the time 80% of the Work as awarded is completed. If the amount payable under any interim bill is not sufficient to cover all deductions to be made for sums advanced with interest and other sums deductible therefrom, the balance outstanding shall be deducted from subsequent interim bill/bills, as may be necessary, failing that, as otherwise provided for in the Contract.
- 23.8 If for any reason, except for reasons of default of the Contractor, the Work under the Contract is suspended continuously for more than fourteen(14) days, the Contractor shall be at liberty to remove the plant and equipment or any part thereof hypothecated to the Employer under GCC Sub-Clause 23.2 above, to any other work site of the Contractor for carrying on his other works, on his furnishing prior to such removal a bank guarantee acceptable to the Employer for the amount of the outstanding advance granted under GCC Sub-Clause 23.2 above with interest and upon the Contractor undertaking to bring back to the Site, before expiry of the period of suspension, such plant and equipment as may be necessary for completion of the Works. If any such Plant and Equipment are not brought back, the Contractor shall forthwith repay the amount for the advance outstanding with interest thereon against such plant and equipment.
- 23.9 The value of the bank guarantees towards any of the above advances shall be reduced every three (3) months after the date of commencement of recovery of corresponding advance and interest thereon under the Contract in accordance with GCC Sub-Clause 23.7. The quantum of reduction of the value of bank guarantee at any point of time shall be proportional to the quantum of work completed as certified by the Engineer-in-Charge **and applicable GST**. The Bank Guarantee value shall stand reduced to zero on completion of total work by the Contractor under the Contract as certified by the Engineer-in-Charge. It should be clearly understood that reduction in the value of bank guarantee towards advance shall not in any way dilute the Contractor's responsibility and liabilities under the Contract including in respect of the Works for which the reduction in the value of bank guarantee is allowed.

24. Payment on Account/Progressive Interim Payments

- 24.1 Bills for progressive payments shall be submitted by the Contractor on monthly basis, on or before the date fixed by the Engineer-in-Charge for the Work executed during the preceding period. The Engineer-in-Charge shall then arrange to have the bill verified for payment.
- 24.2 The progressive payment shall be released after certification by Employer's Field

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Quality Assurance Department that the Works have been performed in accordance with the Technical Specifications and also upon authorisation for the payment by the Engineer-in-Charge. However, the release of first progressive interim payment shall also be subject to submission of documentary evidence by the Contractor towards having taken the insurance policy(ies) in terms of relevant provisions of GCC Clause entitled Contractor's Liability and Insurance' and acceptance of the same by Engineer-in-Charge.

- 24.3 (i) The Contractor shall maintain a separate account with a Scheduled Bank at Site for the purpose of receiving all the payments under the Contract(s) and for utilization of payments received from the Employer for disbursement to sub-contractors, sub-vendors, PRW's etc., of the Contractor. The Contractor shall maintain separate books of accounts for all payments under this Contract and the Engineer-in-Charge shall have access to these at all times.

(ii) Account Tracking Mechanism

In case the Contractor wants to withdraw funds from the above bank account for any purposes other than the Contract, he shall be required to submit an undertaking to the Engineer-in-Charge certifying that all due statutory payments, labour payments and payments to all his sub-contractors have been disbursed corresponding to the total payment received by him under the Contract.

For tracking of payments received from the Employer, the Contractor shall also submit a monthly statement by 7th of every month certifying the transactions pertaining to the above account along with the purpose of such transactions. In case the Contractors transfers funds for any purposes other than the Contract, without prior consent of the Employer, it will constitute a breach of Contract by the Contractor.

- (iii) In case the Contractor violates the above provisions, NTPC will have the right to give suitable instructions to the Bank to regulate / freeze the account.

- 24.4 (i) In case of part acceptance of the Work, the Engineer-in-Charge shall have the right to release payment for that part of the Work.

- (ii) Acceptance of the Work without fulfilling all the obligations mentioned under rates and measurement in Technical Specifications shall be considered as part acceptance of Work.

- 24.5 Payment on account for amount admissible shall be made on the Engineer-in-Charge certifying the sum to which the Contractor is considered entitled by way of interim payment for the following:

- (a) all Work executed, after deducting therefrom the amounts already paid, the performance security deposit and such other amounts as may be deductible

or recoverable in terms of the Contract;

- (b) 75 per cent of the cost (as assessed by the Engineer-in-Charge) of any materials which are in the opinion of the Engineer-in-Charge reasonably required in accordance with the Contract with respect to the period of utilisation & quantum and have been brought to Site for incorporation in the Works and are safeguarded against loss due to any cause whatsoever to the satisfaction of the Engineer-in-Charge, but have not been so incorporated; provided the Contractor provides an insurance cover for the full cost of such materials.
- (c) Unless specifically certified/authorised by the Engineer-in-Charge, no such payment shall be made against the materials brought to Site which in the opinion of the Engineer-in-Charge are not likely to be used/incorporated in the Works in the ensuing period of four (4) months.

24.6 The payments against materials brought to Site under (b) above shall be adjusted as and when materials are utilised/ incorporated in the Works.

24.7 The Contractor shall submit his bill, by the date stipulated by the Engineer-in-Charge, in the prescribed proforma, supported with measurements, jointly acknowledged and accepted in the measurement books. Payments of the Contractor's bill shall be paid by the Employer within twenty-one (21) days from the date of submission of bill subject to the authorisation of the Engineer-in-Charge. Alternatively, if so desired by the Contractor, after preliminary scrutiny and certification by the Engineer-in-Charge, 75% of the certified net payable amount shall be made by the Employer within seven (7) days. The amount certified shall account for all deductions, including statutory deductions as for **GST (if applicable)**, income tax, etc., recoveries for advances and any other amounts due from the Contractor. The balance 25% shall be paid within twenty-one (21) days, from the date of submission of the said bill. Such payments made by the Employer shall not constitute any acceptance of the measurements of items of the Works by the Employer and the Engineer-in-Charge shall have the right to alter, modify, reduce or diminish the quantities or classification entered in the Measurement Books or Bills. The Employer shall have right to recover any amount paid in an earlier bill from any subsequent bill and should the amount to be recovered be more than the amount of the subsequent bill, the Contractor shall on demand from the Engineer-in-Charge or Employer immediately refund the extra amount to the Employer within seven(7) days. Wherever technically feasible, the payments shall be released electronically only as per details of bank account indicated in the Contract.

24.8 Any interim certificate given relating to Work done or materials delivered, may be modified or corrected by any subsequent interim certificate or by the final certificate. No certificate of the Engineer-in-Charge supporting an interim payment shall itself be conclusive evidence that any Work or materials to which it relates is/are in accordance with the Contract.

24.9 In case of the delayed Work beyond the scheduled completion period, pending consideration of extension of time of completion if it had been requested by the Contractor, interim payments shall continue to be made as herein above provided.

24.10 GST applicable upon on account payment shall be paid/reimbursed to the Contractor along with such payments, on production of satisfactory documentary evidence by the Contractor. However, GST as applicable on Advance payment shall be paid to the Contractor along with the Advance sanctioned. The GST paid along with advance shall be adjusted prorata against the tax due upon on account payment, based on the value of the advance recovered from such on account payment.

Notwithstanding anything to contrary contained in the Contract, the Contractor's right to payment under the Contract is subject to issuance of valid tax invoice, payment of applicable GST to the credit of appropriate Government and submission of valid particulars of tax invoice under GST returns in accordance with GST Law.

The Contractor shall issue tax invoices, file appropriate returns, and deposit the applicable GST to the account of appropriate government within the time limit prescribed under the GST Law. In the event of any default, Contractor shall be liable to pay any penalty/demand raised on NTPC due to default by Contractor, and the same shall be recovered/Contractor shall make good the loss.

The Contractor shall be responsible for the issuance of e-way bill and other compliances relating to e-way bill as per GST law.

25. **Contract Price Adjustment**

25.1 The "Contract Price" (for definition please refer to GCC Clause titled "Definition") as awarded shall be the Base Contract Price.

25.2 The rates quoted by the Bidder shall be the base price which will be subject to price adjustment in accordance with the conditions and formula prescribed herein and further subject to satisfying the requirements specified in this clause only.

25.3 A certain fixed percentage of the base price shall not be subject to any price adjustment. The balance percentage to be specified, shall be of identified components towards labour, material(s), steel, cement and Diesel oil/(POL) and shall be subject to Price Adjustment.

25.4.1 The actual amount of Price Adjustment shall be determined by satisfying the conditions specified herein and shall not exceed the maximum limit as specified in SCC.

25.4.2 The value of "F" as specified in these **Special** Conditions of Contract will remain unchanged and fixed component shall not be disputed.

25.5 No ceiling limit on Price Adjustment on the base contract price.

25.6 Price Adjustment(s) shall be calculated for the quantum of works executed for the month or the period of the bill as per agreed Work Schedule. For the purpose of payment/recovery of price variation provisions, the Contractor would be eligible for

such claims or shall be liable for refund on the quantum of work scheduled or the actual quantum of work done provided always that the quantum of work done is more than or equal the scheduled quantum of work as per the agreed Work Schedule. In other words, the Contractor shall not be eligible for these claims nor liable or payment/refund under these provisions if the work has been delayed beyond the scheduled date(s) for reasons attributable to Contractor. However, the Contractor would be eligible for claim or liable for refund for Price Adjustment(s) for quantities of work executed beyond the scheduled dates based on the value of the indices as applicable to the scheduled dates of execution, for such delayed work provided that if the indices during the extended periods are lower than the indices during scheduled dates of execution, then lower indices shall be applicable.

- 25.7.0 Additional, altered or substituted items of work, derived from the agreed schedule of rate (to be attached in the "Award Letter") will also attract price adjustment as per this clause. No price adjustment is payable for the rate/amount of the additional, altered or substituted items of works, when derived from or based on Market rates as per clause 49 of Section IV, GCC. No price adjustment for the cost of Owner Issued materials (viz. steel & cement or any other item(s) issued from Owner's Store) shall be applicable, which are issued at free of cost to the Contractor.
- 25.8.0 Every month after the award of Contract, the Contractor shall submit to the Engineer-in-Charge, a written notice of the changes, if any, that have occurred in the specified indices of materials, labour, Gases or High Speed Diesel Price, during the previous reporting period containing the effective date of such change, the amount of change for the claim of the amount of Contract price adjustment with authenticated documentary evidence of the relevant published indices/diesel price to substantiate the price adjustment.
- 25.9.0 Provided further that such payment/refund shall not be operative and payable after the Schedule expiry of the Contract period or authorized extended Contract period or extended date of completion of works or items of works in question, whichever is earlier.
- 25.10.0 In cases, the work or items of work, or group of items of work, are delayed beyond the schedule dates for the work, for reasons attributable to the Contractor, the price adjustment provisions shall not be applicable for the period of time between the schedule date and the actual date, but for as provided above in sub-clause 25.6.0.
- 25.11.0 For this purpose, the schedule date of work shall be as identified in line with provisions of clause entitled "TIME AND EXTENSION FOR DELAY" and/ or "The Work Schedule/ Bar Chart" (which will be discussed and finalised before Award of work) wherein the separate period of completion has been specified/ agreed to for items, or groups of items, or works.
- 25.12.0 **TOTAL ADJUSTED CONTRACT PRICE**
- The total adjusted Contract price shall be Sigma (ACP) + other elements of Contract price if any.
- 25.13.0 Except as provided therein, no other expenditure incurred by the Contractor, due to levy of additional/increase in royalty, insurance premium(s) benefits to workers/ Labourers or any other Clause(s)/ items(s) due to any reason whatsoever, shall be

payable to the Contractor.

25.14.0 **Tender specifying provisions other than those specified above run the risk of rejection.**

25.15.0 The Contractor shall be required to produce necessary "Price List" for High Speed Diesel at the Indian Oil Corpn. retail outlet nearest to the project and "Monthly Bulletins" issued by office of the Economic Adviser, Ministry of Commerce & Industry, Government of India/ Labour Bureau, Shimla (as published by RBI) or any other related bulletin for materials/ Labour for receiving payments from Employer/ Refund to be made to Employer as required in the Price Adjustment Formula mentioned above.

26. **Taxes, Duties, Levies etc.**

26.1 Except as other wise specifically provided in the Contract, the Contractor shall be liable and responsible for the payment of all taxes, duties, levies and charges imposed on the Contractor, its Sub-Contractors and those imposed on the Contractor's equipment, materials, supplies and services to be used in the performance of the Contract or furnished under the Contract.

26.2 The Award of the Contract is on 'Works Contract' basis. Notwithstanding GCC Sub-Clauses 26.1 above, the Employer shall bear and pay/reimburse to the Contractor, Goods and Services Tax (GST) applicable on the items of Works Contract described in the Schedule of Quantities. However, the taxes, duties & levies as may be applicable on the materials used for Works Contract shall be to the contractor's account and no separate claim in this regard will be entertained by the Employer. Further, in case of any variation in the rates of the GST after the date seven (7) days prior to deadline set for submission of the Techno-Commercial bids, the same shall be paid/ reimbursed to/ recovered from the Contractor subject to submission of documentary evidence.

26.3 It shall be incumbent upon the Contractor to obtain a registration certificate under the GST Law, and other law(s) relating to levy of tax, duty, cess etc. and necessary evidence to this effect shall be furnished by the Contractor to the Employer. If the Contractor intends to engage itself in quarrying or mining of soil/earth, sand, stone/aggregates, metals, minerals or minor minerals required for the Works, as the case may be, it shall obtain necessary permits under the applicable law for such mining or quarrying from the State/Central Government authorities and pay the fee or charges applicable thereto.

26.4 The Contract Price shall be inclusive of any Royalties or Seigniorage Fee or Cess or other charges payable on the quarried or mined metal, minerals, or minor minerals, as the case may be, at the rate(s) prevailing as on seven (7) days prior to the deadline set for submission of bids.

26.4.1 It shall be the responsibility of the Contractor to ensure that the Royalties or Seigniorage Fee or Cess or other charges on the quarried or mined metal, minerals or minor minerals are paid to the statutory authorities.

26.4.2 The component of Royalties or Seigniorage Fee or Cess or other charges, if applicable in a running account bill, shall only be released by the Employer to the Contractor on submission of the following documents in original:

A) In case the Contractor is the primary license holder of the quarry / mines:

- i) Vehicle wise challan / transit permit and proof of payment of royalty, and
- ii) Any other document required as per the relevant Acts/Rules of the concerned state.

B) In case the Contractor is the purchaser of soil/earth, sand, stone/aggregates, metals, minerals or minor minerals:

- i) Purchase voucher and vehicle wise challan / transit permit and proof of payment of royalty, and
- ii) Any other document required as per the relevant Acts/Rules of the concerned state.

26.4.3 Final bill payment shall be released to the Contractor on production of royalty clearance certificate issued by the concerned statutory authorities for total quantity of quarried or mined metal, minerals, or minor minerals required for the Works.

26.4.4 The Contractor shall pay and indemnify the Employer against any default in payment of Royalties or Seigniorage Fee or Cess or other charges by the Contractor or the agency from which the Contractor purchases soil/earth, sand, stone/aggregates, metals, minerals or minor minerals.

26.4.5 In the event of there being a statutory increase in the rates of royalty charges/fresh levy of royalty on materials, the same shall be reimbursed to the Contractor upon submission of original challan by him of having made the payments at revised rates. In the event of there being a decrease in such rates, the same shall be recovered from the Contractor. The base date for calculating the increase or decrease shall be the rate as on seven (7) days prior to the date of Techno-Commercial bid opening. The total reimbursement (positive or negative) as specified above, to be paid or recovered, shall however be calculated on the quantity of materials actually considered while making the royalty payments to the concerned authorities, or the theoretical consumption of these materials (calculated on the basis of the volume of concrete or fill accepted for payment), whichever is less, and on the basis of documentary evidence of Govt. Notification. However, the Contractor will settle claims, if any, on account of over charge by the State Authorities.

26.5 If a new tax, duty or levy is imposed under statute or law in India after the date seven (7) days prior to deadline set for submission of the bids and the Contractor becomes liable there under to pay and actually pays the said new tax, duty or levy for bonafide use on the Works contracted, the same shall be reimbursed to the Contractor against documentary evidence of proof of payment, provided that the amount thus claimed is not paid/payable under price variation provision of the Contract.

26.6 The payment/reimbursement of statutory variations in the rates of tax and/or of new tax, duty or levy imposed under statute or law in India as per GCC Sub-Clauses 26.2, 26.3, 26.4 and 26.5 above, would be restricted only to direct transactions between the Employer and the Contractor.

26.7 The Employer shall be entitled to make necessary tax deductions at source as per the prevalent laws. The Contractor shall be required to submit the PAN details to the Engineer-in-Charge before the submission of the first bill/ invoice under the Contract.

26.8 The Contractor shall himself be informed of all the applicable laws, notifications, rules, circulars and other communications of the State or Central or other authorities with regard to levy of any tax, duty, cess, levy or fee etc, which in any manner may impinge upon him in performance of any obligations/responsibilities under or arising out of the Contract.

27. Overpayments and Underpayments

27.1 Wherever any claim for the payment of a sum of money to the Employer arises out of or under this Contract against the Contractor, the Contractor upon demand by the Employer or by the Engineer-in-Charge on behalf of the Employer, with explanation of the reasons for such a sum/ claim becoming due, shall forthwith pay the same to the Employer. If the Contractor fails to do so within twenty-one (21) days of such a claim, then the same may be deducted by the Employer from any sum then due or which at any time thereafter may become due to the Contractor under this Contract or from any other sum due to the Contractor from the Employer which may be available with the Employer or from his security deposit.

27.2 The Employer reserves the right to carry out post payment audit and technical examination of the final bill including all supporting vouchers, abstracts, etc. The Employer further reserves the right to enforce and recover any overpayment when detected, notwithstanding the fact that the amount of the final bill may include any item which is under dispute between the parties and referred to for settlement under GCC Clause entitled 'Settlement of Disputes' and notwithstanding the fact that the amount of the final bill figures in the arbitration decision/award.

27.3 If as a result of such audit and technical examination, any overpayment is discovered in respect of any Work done by the Contractor or alleged to have been done by him under the Contract, it shall be recovered by the Employer from the Contractor by any or all of the methods prescribed above. Similarly, if any underpayment is discovered by the Employer, the amount shall be duly paid to the Contractor by the Employer forthwith.

27.4 Provided that the aforesaid right of the Employer to adjust overpayments against amounts due to the Contractor under any other Contract with the Employer shall not extend beyond the period of two years from the date of payment of the final bill or in case the final bill is a minus bill, from the date the amount payable by the Contractor under the minus final bill is communicated to the Contractor.

27.5 Any amount due to the Contractor under this Contract for underpayment may also be adjusted against any amount then due or which may at any time thereafter become due from the Employer to the Contractor under any other Contract or account whatsoever.

28. Time Limit for submission & payment of Final Bill and waiver of rights of all

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claims

- 28.1 The final bill shall be submitted by the Contractor within three (3) months of physical completion of the Works unless otherwise a longer period is agreed to between the Engineer-in-Charge and the Contractor. No further claims shall be made by the Contractor after submission of the final bill and these shall be deemed to have been waived and extinguished. Payment of the final bill will be made within four (4) months of receipt of the same

D. Execution of Facilities

29. Work Commencement, Execution & Delays

29.1 Commencement of Works

The execution of the Works shall commence from the 14th day after the date on which the Engineer-in-Charge issues written orders to commence the Work, unless otherwise stated elsewhere in the Contract.

29.2 Time for Completion

The entire scope of Work covered under this Contract shall be completed within the time stated in SCC or within such extended time granted to the Contractor by the Employer under the provisions of GCC Sub-Clause 29.5. The time allowed for execution of the Works as specified in the SCC or the extended time in accordance with these Conditions shall be the essence of the Contract.

29.3 Work Progress

- 29.3.1 Unless already incorporated in the Letter of Award, as soon as possible after the Contract is awarded, the Engineer-in-Charge and the Contractor shall agree upon a Work Schedule which will become the Contract Work Schedule. The Work Schedule shall be prepared in direct relation to the time stated in the Contract documents for the completion of the Works. The Work Schedule shall indicate the forecast of the dates of commencement and completion of various trades or sections of work.

- 29.3.2 All the Contractor's activities shall be performed and completed strictly in accordance with the agreed Work Schedule and to achieve the targets, the Contractor shall have to plan adequate mobilisation of all resources. The Engineer-in-Charge, shall however, have the right to review the progress and modify the sequence of carrying out the Work suiting the Site conditions and the Contractor shall be required to comply with such modifications and complete his activities in accordance thereof without any extra cost to the Employer.

29.3.3 Maintenance of Records of Weekly Progress Review Meetings at Site

The Contractor shall be required to attend all weekly site progress review meetings

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organized by the 'Project Manager' or his authorised representative. The deliberations in the meetings shall inter alia include the weekly program, progress of work (including details of manpower, tools and plants deployed by the contractor vis-a-vis agreed schedule), inputs to be provided by Employer, delays, if any and recovery program, specific hindrances to work and work instructions by Employer. The minutes of the weekly meetings shall be recorded in triplicate in a numbered register available with the 'Project Manager', or his authorized representative. These recordings shall be jointly signed by the Project Manager or his authorized representative and the Contractor and one copy of the signed records shall be handed over to the Contractor. The following documents shall form the principal basis for consideration of Time Extension Pursuant to GCC Clause 29.5 with or without LD, determining the compensation amount pursuant to GCC Clause 29.6 and settlement of extra claims during the execution of contract: 1. The joint recordings in the weekly review meetings register 2. Records of Technical Coordination Meetings 3. Records of Contract Review Meetings 4. Written notices issued by the "Engineer in Charge" or his authorized representative to contractor in the relevant period.

29.4 Contract Coordination Procedures, Coordination Meetings & Progress Reporting

29.4.1 The Contractor shall prepare and finalise in consultation with the Engineer-in-Charge, a detailed contract coordination procedure within twenty-eight (28) days from the date of issue of Letter of Award, for the purpose of execution of the Contract.

29.4.2 The Contractor shall have to attend all the meetings at his own cost with Engineer-in-Charge or any authorised representative of the Employer during the currency of the Contract, as and when required and fully cooperate with such persons and agencies involved during these discussions.

29.4.3 During the execution of the Work, the Contractor shall submit at his own cost a detailed monthly progress report to the Engineer-in-Charge in three copies, latest by 14th of every month.

29.5 Extension of Time for Completion

29.5.1 The Time for Completion specified shall be extended if the Contractor is delayed or impeded in the performance of any of the obligations under the Contract by reason of any of the following:

- a) any occurrence of Force majeure as provided in GCC Clause entitled 'Force Majeure', or
- b) Work Schedules for beyond deviation limits & Extra Items as provided in GCC Sub-Clause 49.5, or
- c) any default or breach of the Contract by the Employer, or delay on the part of other contractors engaged by the Employer in executing work not forming part of this Contract, or

- d) any suspension order given by the Employer under GCC Sub-Clauses 50.1 (ii) and 50.1 (iii), or
- e) Any other sufficient cause which, in the opinion of the Engineer-in-Charge, is beyond the Contractor's reasonable control;

by such period as shall be fair and reasonable in all the circumstances and as shall fairly reflect the delay or impediment sustained by the Contractor.

29.5.2 Except where otherwise specifically provided in the Contract, the Contractor shall submit to the Employer a notice in writing of a claim for an extension of the Time for Completion, together with particulars of the event or circumstance justifying such extension as soon as reasonably practicable, but no later than twenty-eight (28) days after the commencement of such event or circumstance. As soon as reasonably practicable, after the receipt of such notice and supporting particulars of the claim, the Employer shall give a fair and reasonable extension of time for completion of Work. Such extension shall be communicated to the Contractor by the Engineer-in-Charge in writing, within fifty-six (56) days of the date of receipt of such request by the Engineer-in-Charge.

29.5.3 The Contractor shall at all times use his reasonable efforts to minimise any delay in the performance of his obligations under the Contract.

29.5.4 The compensations, if any, payable to the Contractor on account of any one or more of the above reasons of delay have been separately dealt with under relevant provisions of the Contract.

29.6 **Liquidated Damages for Delay**

29.6.1 If the Contractor fails to complete the Work on or before the scheduled or extended date of completion as per GCC Sub-Clauses 29.2 and 29.5, he shall, without prejudice to any other right or remedy of the Employer, arising out of the Contract on account of such delay, be liable for payment of liquidated damages, not as penalty, as per provisions of SCC Clause entitled 'Liquidated Damages for Delay'.

29.6.2 The following documents shall form the principal basis for consideration of Extension of Time for Completion pursuant to GCC Sub-Clause 29.5 with or without Liquidated Damages and determining the compensation amount pursuant to GCC Sub-Clause 29.6.

1. The joint recordings in the periodic meeting register,
2. Records of Technical Coordination meetings,
3. Records of Contract Review meetings,
4. Written notices issued by the Employer and/or the Engineer-in-Charge or his authorized representative to the Contractor in the relevant period.
5. Written requests/ notices by the Contractor to Employer/ Engineer-in-Charge in

the relevant period.

29.7 Delays by Employer or his Authorised Representative

29.7.1 In case the Contractor's performance is delayed due to any act of omission on the part of the Employer or his authorised representative, then the Contractor shall be given appropriate extension of time for the completion of the Works, to the extent such omission on the part of the Employer has caused delay in the Contractor's performance of his work. Regarding reasonableness or otherwise of the extension of time, the decision of the Engineer-in-Charge shall be final.

29.7.2 If such delays by the Employer have resulted in any increase in the cost to the Contractor, the Contractor shall be eligible to claim demonstrable and reasonable costs supported by full details of such increased costs incurred by him with all documentary evidence. The Employer shall examine the justification for such a request for claim and if satisfied, the extent of compensation shall be mutually agreed depending upon the circumstances at the time of such an occurrence.

30. Sub Contracts

30.1 After the award of the Contract, the Contractor shall not subcontract the Works/ any part of the Works without the prior written consent of the Engineer-in-Charge. Any such consent shall not relieve the Contractor from any liability or obligation under the Contract and he shall be responsible for the acts, defaults and neglects of any Sub-Contractor, his agents, servants or workmen as fully as if they were the acts, defaults or neglects of the Contractor, his agents, servants or workmen.

Provided that the Contractor shall not be required to obtain such consent for:

- a) the provision of labour,
- b) the purchase of materials which are in accordance with the standards specified in the Contract, or
- c) the subcontracting of any part of the Works for which the sub-contractor is named in the Contract.

In the event of the Contractor proposing a sub-contractor for any part of the Works after the award of the Contract, he shall be required to take approval from the Engineer-in-Charge. If the Engineer-in-Charge approves of the same, he shall stipulate qualifying requirements for the sub-contractors to perform the specified part of Work. The Contractor will then submit the requisite credentials of the agency(ies) he proposes to engage. The details so furnished by the Contractor shall be reviewed by the Employer. In case the agency(ies) proposed by the Contractor for the subcontracting are not considered acceptable, the Contractor will be required to furnish credentials of alternate agency(ies) for approval of Engineer-

in-Charge. Based on the review and assessment, the agency(ies) shall be approved by the Engineer-in-Charge within twenty-one (21) days of furnishing of credentials by the Contractor.

- 30.2 Where a list of approved agencies for a sub-contracting work is provided in the Contract, the Contractor shall inform the name of the sub-contractor selected by him within a period as agreed with the Engineer-in-Charge, however not later than twenty-eight (28) days of the date of such selection.

31. Setting out the Works

- 31.1 The Engineer-in-Charge shall supply dimensioned drawings, levels and other information necessary to enable the Contractor to set out the Works and the Contractor shall set out the Works and be responsible for the accuracy of the same. He shall amend at his own cost and to the satisfaction of the Engineer-in-Charge any error found at any stage which may arise through inaccurate setting out unless such error is based on incorrect data furnished in writing by the Engineer-in-Charge, in which case the cost of rectification shall be borne by the Employer. The Contractor shall protect and preserve all bench marks used in setting out the Works till date of submission of final bill under the Contract, unless the Engineer-in-Charge directs otherwise.

32. Methodology of Construction & Equipment Mobilisation

- 32.1 Methodology of construction and the work plan adopted by Contractor shall match the construction methodology/requirements specified in Technical Specifications.
- 32.2 The suggested minimum plant & equipment and machinery to be deployed by the Contractor for the execution of Work shall be as given in Technical Specifications.
- 32.3 The Contractor shall arrange at his own expense all tools, plant and equipment including Crane(s) (hereinafter referred to as T&P) required for execution of the work.
- 32.3.1 If the Contractor requires any item of T & P on hire from the Employer, the Employer will, if such item is available, hire it to the Contractor at an hourly rate to be fixed by the Engineer-in-Charge.
- 32.3.2 The T&P shall be given to the Contractor on hire by the Employer for a period of one hour or its multiple thereof. In case the T&P is hired by the Contractor for a period of four hours or less, the hire charges applicable for a minimum period of four hours shall be recovered from the Contractor's bills. In case the T&P is hired by the Contractor for a period exceeding four hours, the hire charges shall be calculated based on the charges applicable as per hourly rate. The hire charges in respect of T&P given on hire to the Contractor by the Employer shall be recovered from the Contractor's bills.
- 32.3.3 For accounting purpose, total working hours shall be considered to be the period between time of placement of T&P to the Contractor at the requested location and time of release of the same. This shall be logged in Record Book on daily basis

and shall be signed between Contractor/ Subcontractor and the Engineer-in-charge or his authorised representative. In case the T&P issued to the Contractor is not owned by the Employer but hired from another agency, the authorised representative of the agency providing the T&P will also sign the said Record Book. In case the Contractor contests correctness of any entry and/or fails to sign the Record Book the decision of the Engineer-in-Charge shall be final and binding on him. Hire charges shall be calculated in accordance with the entries in the Record Book.

32.3.4 The Contractor will be exempted from levy of any charges for the number of days he is called upon in writing by the Engineer - in- Charge to suspend execution of the work, provided Employer's T & P/ T&P hired by the Employer in question has, in fact, remained idle with the Contractor because of the suspension, provided the Contractor, in case the period of suspension exceeds 11 days, returns Employer's T & P/ T&P hired by the Employer to the place from where it was issued.

32.3.5.1 **T&P owned by Employer**

The hire charges will cover financing cost, charges of crew, depreciation, stores for maintenance and cleaning purposes and fuel needed to start a machine at the time of issue. All other charges such as cost of fuel for running a machine, engine oil, kerosene oil, etc., for working Employer's T&P, and all unskilled labour and water required for servicing/wash out shall be borne by the Contractor. The Contractor shall permit the Engineer-in-Charge to carry out periodical maintenance of Employer's T&P in accordance with the provision there for in the aforesaid Schedule, and there will be no deduction in hire charges for the period spent on such maintenance.

However, the Contractor shall be allowed to return the tools and plants (issued by the Employer) for purposes of repairs and for the duration of such repairs no hire charges shall, be levied.

The Contractor shall be responsible for care and custody of Employer's T&P (including employment of chowkidars) during the period Employer's T&P remain with him and any damage (fair wear and tear excepted) to any of the equipment(except for Excepted Risks provided always the Contractor has taken precautions necessary to protect it from such risks) shall be made good at the Contractor's expense to the satisfaction of the Engineer-in-Charge unless such damage is caused because of negligence of crew provided by the Employer.

32.3.5.2 **T&P hired by the Employer**

The hourly hiring rate for T&P hired by the Employer from another agency and issued to Contractor shall be all inclusive rate including the cost of hiring, operation & maintenance charges, fuel charges and other charges.

32.3.6 The Employer gives no guarantee in respect of output of T & P hired to the Contractor and no reduction in rates or any compensation shall be allowed on the ground that out turn or performance of Employer's T & P/ T&P hired by the

Employer was not to the Contractor's expectation.

- 32.3.7 The T&P hired to the Contractor shall be returned at the place of issue by the Contractor to the Engineer-in-Charge (unless otherwise directed) on execution of the work or section of the work at the end of the day. In case the T&P is used by the Contractor in continuation of previous requisition and the crane has not moved out of his work area, then the movement of crane for fresh requisition(s) by the Contractor within his work area shall be to the Contractor's account.
- 32.3.8 The Employer shall be entitled to terminate the hire without assigning any reason whatsoever and the Contractor shall have no claim to any payment of compensation or otherwise whatsoever on account of termination of hire of T&P issued by the Employer. On termination of the hire by the Employer, the Contractor shall return the T&P at the place of issue unless otherwise directed by the Engineer-in-Charge.
- 32.4. The Contractor shall arrange at his own expense all tools, plant and equipment including Crane(s) (hereinafter referred to as T&P) required for execution of the work.
- 32.4.1 If the Contractor requires any item of T&P on hire from the Corporation, the corporation will, if such item is available, hire it to the Contractor at an hourly rate to be fixed by the Engineer-in-Charge.
- 32.4.2 The T&P shall be given to the Contractor on hire by the Corporation for a period of one hour or its multiple thereof. In case the T&P is hired by the Contractor for a period of four hours or less, the hire charges applicable for a minimum period of four hours shall be recovered from the Contractor's bills. In case the T&P is hired by the Contractor for a period exceeding four hours, the hire charges shall be calculated based on the charges applicable as per hourly rate. The hire charges in respect of T&P given on hire to the Contractor by the Corporation shall be recovered from the Contractor's bills.
- 32.4.3 For accounting purpose, total working hours shall be considered to be the period between time of placement of T&P to the Contractor at the requested location and time of release of the same. This shall be logged in Record Book on daily basis and shall be signed between Contractor/ Subcontractor and the Engineer-in-charge or his authorised representative. In case the T&P issued to the Contractor is not owned by the Corporation but hired from another agency, the authorised representative of the agency providing the T&P will also sign the said Record Book. In case the Contractor contests correctness of any entry and/or fails to sign the Record Book the decision of the Engineer-in-Charge shall be final and binding on him. Hire charges shall be calculated in accordance with the entries in the Record Book.
- 32.4.4 The Contractor will be exempted from levy of any charges for the number of days he is called upon in writing by the Engineer-in-Charge to suspend execution of the work, provided Corporation's T&P/T&P hired by the Corporation in question has, in fact, remained idle with the Contractor because of the suspension, provided the Contractor, in case the period of suspension exceeds 11 days, returns

Corporation's T&P/T&P hired by the Corporation to the place from where it was issued.

33. Patent Indemnity

33.1 The Contractor shall indemnify and hold harmless the Employer and its employees and officers from and against any and all suits, actions or administrative proceedings, claims, demands, losses, damages, costs, and expenses of whatsoever nature, including attorney's fees and expenses, which the Employer may suffer as a result of any infringement or alleged infringement of any patent, utility model, registered design, trademark, copyright or other intellectual property right registered or otherwise existing at the date of the Contract in the performance of the Contract.

33.2 In the event of any claim being made or action being brought against the Employer or its representatives or its employees, in respect of any such matters as aforesaid, the Contractor shall immediately be notified thereof. However, such indemnity shall not apply when such infringement has taken place in complying with the specific directions issued by the Employer; but the Contractor shall pay any royalties or other charges payable in respect of any such use, the amount so paid being reimbursed to the Contractor only if the use was as a result of any drawings and/or specifications issued after the award of Contract by the Employer, provided further that the Contractor has brought to the notice of the Engineer-in-Charge, of such infringement immediately upon the instructions of the Engineer-in-Charge or upon the Contractor becoming aware of such infringement.

34. Materials for the performance of the Contract

(a) Materials to be provided by the Contractor

1. The Contractor shall at his own expense, provide all materials required for the Works other than those which are to be issued by the Employer.
2. All materials to be provided by the Contractor shall be in conformity with the specifications laid down in the Contract and the Contractor shall, if requested by the Engineer-in-Charge, furnish proof to the satisfaction of Engineer-in-Charge that the materials so comply.
3. Wherever required by the Engineer-in-Charge, the Contractor shall, at his own expense and without delay, provide samples of materials proposed to be used in the Works. The Engineer-in-Charge shall within seven (7) days thereafter or within such further period as he may require, intimate to the Contractor in writing, whether samples are approved by him or not. If samples are not approved, the Contractor shall forthwith arrange for fresh samples complying with the Technical specifications laid down in the Contract, for approval.
4. The Engineer-in-Charge shall have full powers to require removal of any or all of the materials brought to site by the Contractor which are not in accordance with the Contract specifications or do not conform in character or quality to samples approved by him. In case of default on the part of the Contractor in

removing rejected materials, the Engineer-in-Charge shall be at liberty to have them removed by other means. The Engineer-in-Charge shall have full powers to order the Contractor to provide other proper materials to be substituted for rejected materials and in the event of the Contractor refusing to comply, he may cause the same to be supplied by other means. All costs, which may attend upon such removal and/or substitution, shall be borne by the Contractor.

5. The Engineer-in-Charge shall be entitled to have tests carried out as specified in the Contract for any materials supplied by the Contractor other than those for which, as stated above, satisfactory proof has already been furnished, at the cost of the Contractor and the Contractor shall provide at his expense all facilities which the Engineer-in-Charge may reasonably require for the purpose. If no tests are specified in the Contract, and such tests are required by the Engineer-in-Charge, the Contractor shall provide all facilities required for the purpose and the charges for these tests including the cost of materials consumed/used in such tests shall be to the account of Employer, except if the tests disclose that the said materials are not in accordance with the provision of the Contract, then the same shall be to the account of the Contractor.
6. The Contractor shall indemnify the Employer, its representatives or its employees against any action, claim or proceeding relating to infringement or use of any patent or design or any alleged patent or design rights and shall pay any royalties or other charges which may be payable in respect of any article or material or part thereof included in the Contract. In the event of any claim being made or action being brought against the Employer or its representatives or its employees, in respect of any such matters as aforesaid, the Contractor shall immediately be notified thereof. However, such indemnity shall not apply when
such infringement has taken place in complying with the specific directions issued by the Employer; but the Contractor shall pay any royalties or other charges payable in respect of any such use, the amount so paid being reimbursed to the Contractor only if the use was as a result of any drawings and/or specifications issued after the award of Contract by the Employer, provided further that the Contractor has brought to the notice of the Engineer-in-Charge, of such infringement immediately upon the instructions of the Engineer-in-Charge or upon the Contractor becoming aware of such infringement.
7. Subject as hereinafter provided in GCC Clause entitled 'Contract Price Adjustment' all charges on account of GST and other levies/octroi on materials obtained for the Works from any source (excluding materials issued by the Employer) shall be borne by the Contractor.

(b) **Materials to be issued by the Employer:**

(i) **Materials to be issued by the Employer free of cost**

The Employer, if so stipulated in SCC, may issue cement, reinforcement steel and structural steel to the Contractor free of cost for incorporation in the Works as per the terms and conditions specified in the SCC/Technical

Specifications.

(ii) Materials to be issued by the Employer on chargeable basis

If after the award of the Contract, the Contractor desires the Employer to issue/supply any other materials, for the purposes of the Contract such materials may be issued by the Employer, if available, at rates and terms and conditions to be fixed by the Engineer-in-Charge. The Employer reserves the right not to issue any such materials. The non-issue of such materials will not entitle the Contractor for any compensation whatsoever either in time or in cost.

(c) General:

1. Materials required for the Works, whether brought by the Contractor or issued by the Employer, shall be stored by the Contractor only at places approved by the Engineer-in-Charge. Storage and safe custody of material shall be the responsibility of the Contractor.
2. Engineer-in-Charge shall be entitled at any time to inspect and examine any materials intended to be used in or on the Works, either on the Site or at factory or workshop or other place(s) where such materials are assembled, fabricated, manufactured or at any place(s) where these are lying or from which these are being obtained and the Contractor shall give such facilities as may be reasonably required for such inspection and examination.
3. All materials brought to the Site shall not be removed off the Site without the prior written approval of the Engineer-in-Charge. But whenever the Works are finally completed and advance, if any, in respect of any such material is fully recovered, the Contractor shall at his own expense forthwith remove from the Site all surplus material originally supplied by him.
4. The Employer may issue all the materials agreed to be issued to the Contractor under the Contract, at its site stores, or nearest railhead. In case the materials are issued at the nearest rail head, the cost of transportation only, from such rail head to the Site will be borne by the Employer, subject to the reasonableness of such transportation cost being certified by the Engineer-in-Charge. All other costs such as loading, unloading, transportation to Contractor's godown, storage etc. till the materials are incorporated in the Works or returned to the Employer shall be to the account of the Contractor.
5. All materials issued to the Contractor, by the Employer for incorporation or fixing in the Works (including preparatory work) shall, on completion or on termination of the Contract, be returned by the Contractor at his expense, at the Employer's store, after making due allowance for actual consumption, reasonable wear and tear and/or waste. If the Contractor is required to deliver such materials at a place other than the Employer's

store, he shall do so and the transportation charges from the Site to such place, less the transportation charges which would have been incurred by the Contractor, had such materials been delivered at the Employer's store, shall be borne by the Employer.

35. Quality Assurance Programme

- 35.1 Sampling, testing and quality assurance requirements are given in Technical Specifications.
- 35.2 All costs associated with testing of materials required as per Technical Specifications shall be deemed to be included in Contract rates/prices in the Schedule of Quantities.

36. Inspection and Approval

- 36.1 All Works embracing more than one process shall be subject to examination and approval at each stage thereof and the Contractor shall give due notice to the Engineer-in-Charge or his authorised representative when each stage is ready. In default of such notice, the Engineer-in-Charge shall be entitled to appraise the quality and extent thereof.
- 36.2 No work shall be covered up or put out of view without the approval of the Engineer-in-Charge or his authorised representative and the Contractor shall provide full opportunity for examination and measurement of any work which is about to be covered up or put out of view and for examination of foundations before further work is placed thereon. The Contractor shall give due notice to the Engineer-in-Charge or his authorised representative whenever any such work or foundation is ready for examination and the Engineer-in-Charge or his representative shall without unreasonable delay, unless he considers it unnecessary and advises the Contractor accordingly' attend for the purpose of examining and measuring such work or of examining such foundations. In the event of the failure of the Contractor to give such notice he shall, if required by the Engineer-in-Charge, uncover such work at his own expense.
- 36.3 The Engineer-in-Charge or his authorised representative shall have powers at any time to inspect and examine any part of the Works and the Contractor shall give such facilities as may be reasonably required for such inspection and examination.
- 36.4 The Contractor shall uncover any part of the Works and/or make openings in or through the same as the Engineer-in-Charge may from time to time direct for his verification and shall reinstate and make good such part to the satisfaction of the Engineer-in-Charge. If any such part has been covered up or put out of view after being approved by the Engineer-in-Charge and is subsequently found on uncovering to have been executed in accordance with the Contract, the expenses of uncovering and/or making opening in or through, reinstating and making good the same shall be borne by the Employer. In any other case all such expenses shall be borne by the Contractor.
- 36.5 The additional & specific inspection and approval requirements in respect of the

Works are detailed further in the Technical Specifications.

37. Records and Measurement

- 37.1 The Engineer-in-Charge shall, except as otherwise stated, ascertain and determine by measurement the value of the Work done in accordance with the Contract.
- 37.2 All items having a financial value shall be entered in Measurement Book, level book, etc. prescribed by the Engineer-in-Charge so that a complete record is obtained of all Work performed under the Contract.
- 37.3 Measurements shall be taken jointly by the Engineer-in-Charge or his authorised representative and the Contractor or his authorised representative.
- 37.4 Before taking measurements of any Work the Engineer-in-Charge or his authorised representative for the purpose shall give a reasonable notice to the Contractor. If the Contractor fails to attend or send his authorised representative for taking the measurements after such a notice or fails to countersign or to record the objection, if any, within a week from the date of measurement, then in any such event measurements taken by the Engineer-in-Charge or his authorised representative shall be taken to be correct measurements of the Work.
- 37.5 The Contractor shall, without extra charge, provide assistance with every appliance, labour etc. necessary for taking measurements.
- 37.6 Measurements shall be signed and dated by both parties each day on the Site on completion of measurement. If the Contractor objects to any of the measurements recorded, a note to that effect shall be made in the Measurement Book against the item objected to and such note shall be signed and dated by both parties engaged in taking the measurement. The decision of the Engineer-in-Charge on any such dispute or difference or interpretation shall be final and binding on both the parties and shall be beyond the scope of the provisions of settlement of disputes under the Contract.

38. Methods of Measurement

- 38.1 Measurement of Contract items of Work shall be taken in accordance with method of Measurement stipulated in the Technical Specifications/Schedule of Quantities. In case of extra items, the Engineer-in-Charge shall also specify the method of measurement for such items at the time of his order for execution of such extra items.
- 38.2 In case no method of measurement is stipulated in Technical Specifications / Schedule of Quantities/ Order of the Engineer-in-Charge, then the Method of Measurement of such items shall be as per the relevant Standard Method of Measurement issued by Indian Standards Institution or general industry practice/ local custom.

39. Temporary / Enabling Works

39.1 The siting and nature of all offices, access road to the work areas, access tracks to work areas, sumps, and all other Temporary / Enabling Works as may be required for the proper execution of the Works shall be subject to the approval of the Engineer-in-Charge. These Works shall be executed by the Contractor at his own cost. Hard crusting for pre-assembly/fabrication yard shall be in line with Technical Specifications.

39.2 All equipment, labour, materials including cement, reinforcement and the structural steel required for the Enabling Works associated with the entire Contract shall have to be arranged by the Contractor only. Nothing extra shall be paid to the Contractor on this account and the unit rates quoted by the Contractor on this account and the unit rates quoted by the Contractor for various items in the Schedule of Quantities shall be deemed to include the cost of Enabling Works.

39.3 However, for fabrication yard or for fabrication of structural steel, if any, hard crusting made with compacted filling using broken hard stone aggregate with binding material shall be measured and paid under relevant item of stone aggregate filling with binding material as specified in Technical Specifications and Schedule of Quantities.

The maximum area of hard crusting that will be paid is limited to consecutive three (3) month peak fabrication quantity in M.T. indicated in the Work Schedule multiplied by 3.5 sq. m per M.T. The hard crusting area arrived as above shall be further subject to availability of appropriate area in the general layout plan and approval of the Engineer-in-Charge.

Further development of fabrication and assembly bed, power distribution points, cable laying, drains, additional area of hard crusting over and above area stipulated etc., shall be done by the Contractor, at his own cost.

39.4 The Contractor shall make his own arrangement for approach to the work Site, including borrow / disposal area and for movement of men, machinery, other requirement etc. required for carrying out the Work included under this Contract.

40. **Urgent Works**

40.1 If any Urgent Work becomes necessary and the Contractor is unable or unwilling at once to carry it out, the Engineer-in-Charge may by his own or other means, carry it out as he may consider necessary. If the Urgent Work shall be such as the Contractor is liable under the Contract to carry out at his expenses all expenses, incurred on it by the Employer shall be recoverable from the Contractor and be adjusted or set off against any sum payable to him.

41. **Construction Power and Construction Water Supply and Staff / Labour Colony**

Unless otherwise stated in SCC, the provisions for Construction Power and Construction Water Supply and Staff / Labour Colony shall be as under: 41.1.1 The Contractor shall advise the Engineer-in-Charge, within twenty-eight (28) days from the date of acceptance of the Letter of Award, about his exact requirement of space for his office, storage area, preassembly and fabrication areas, labour and

staff colony area (land if available), etc. The above requirement shall be reviewed by the Engineer-in-Charge and space as decided by him will be allotted for his use as well as his Sub-Contractor's use.

41.1.2 On completion of Work, the Contractor shall handover the land duly cleaned to the Engineer-in-Charge. Until and unless the Contractor has handed over the vacant possession of land allotted to him for the above purpose, the payment of his final bill shall not be made. The Contractor shall be made liable to pay for the use and occupation at the rates to be determined by the Engineer-in-Charge if the Contractor overstay in the land after the Contract is completed.

41.1.3 The Contractor shall submit to the Engineer-in-Charge within twenty-eight (28) days from the date of acceptance of the Letter of Award, his electrical power requirements, if any, to allow the planning of the same by the Engineer-in-Charge. The Contractor shall be provided with free supply of electricity for the purposes of the Contract only, at two convenient locations in the Site. The Contractor shall make his own further distribution arrangement. All temporary wiring must comply with local regulations and will be subject to Engineer-in-Charge's inspection and approval before connection to supply. The free supply of power will not be provided for the use in the labour and staff colony. Power supply for labour and staff colony shall be provided at one point. It shall be the responsibility of the Contractor to take the power supply upto the point of his use. The Contractor shall be charged for power supply to labour and staff colony at rates prevalent as per the tariff of the electricity distribution entity at the Site.

The Engineer-in-Charge may consider additional points of power supply in deserving and exceptional cases.

41.1.4 The Employer does not guarantee uninterrupted power supply.

41.1.5 The Contractor shall arrange for drinking water to his workmen/staff at Work Site and other water supply for all purposes for his labour and other personnel at the worksite / colony on his own. The quality of water should meet the requirements for which it is proposed to be used. All Civil and Structural Work associated with the above including borings, pipe lines, valves, pumps, tube wells, pump house, underground storage tank, over ground storage tank, water tankers etc., whatsoever required for taking the water from the underground source or any other source to the place of use shall be provided / erected/ constructed / maintained by the Contractor at his own cost.

41.1.6 The Contractor shall not be entitled to any compensation on account of the expenditure incurred in arranging the construction water.

41.2 Supply of Unfiltered Water for Construction Purposes only

41.2.1 Unless otherwise stated in SCC, the Contractor shall draw water from the water supply mains provided in the project at suitable points to be indicated by the Engineer-in-Charge. All pipe lines, pumps and other accessories required for taking the water from the mains to the site of Work shall be provided by the Contractor at his own cost. He shall not be entitled to any payment on account of the expenditure

incurred in providing the pipe lines, pumps, etc. No charges will be levied on the Contractor for the water drawn by him for the purpose of the construction work. Should the water, however, be used for either the colony or for manufacture, always subject to prior written permission of the Engineer-in-Charge, the same shall be chargeable at a rate to be fixed by him whose decision in this regard shall be final. However, the water supply shall have to be taken through a metered connection.

- 41.2.2 The Employer does not guarantee the maintenance of uninterrupted supply of water and in case of any interruptions of such supply of water; the Contractor shall be responsible for making at his own cost alternative arrangements for water. The Engineer-in-Charge also reserves the right to limit the quantity of water to be allowed to be drawn by the Contractor.
- 41.2.3 No claim for damages will be entertained by the Employer on account of interruption of water supply or limitation of quantity of water as aforesaid or on account of the water so supplied being not fit for construction purposes or on any other account in connection with such water supply.
- 41.2.4 It will be the responsibility of the Contractor to adequately treat the water at his cost before use for the intended purpose.
- 41.2.5 Where the Contractor makes his own arrangements for water required for the Work, nothing extra shall be paid for the same. He should make arrangements for storage of sufficient quantity of water required for at least a day's work.
- 41.3 The contractor shall arrange for construction water from underground/local sources. All borings, pipe lines, pumps, water tankers, underground storage tank, over-ground storage tank, etc, whatsoever required for taking the water from the underground source to the site of work shall be provided / erected / constructed / maintained by the contractor at his own cost.
- 41.4 The Contractor shall not be entitled to any compensation on account of the expenditure incurred in arranging the construction water.
- 41.5 Contractor shall make the arrangement for construction power at his own cost. However, NTPC may provide construction power as per site availability on chargeable basis, but the Contractor shall not be entitled to any compensation on account of the quality and availability of power from NTPC.
- 41.6 The contractor shall make his own arrangement for construction/ rented premises for labour / staff colony.
42. **Site Laboratory**
- 42.1 As part of the Contract, the Contractor shall provide and maintain a site laboratory for the testing of construction materials under the direction and general supervision of the Engineer-in-Charge.

- 42.2 The laboratory building shall be constructed and installed with the appropriate facilities. Temperature and humidity controls shall be available wherever necessary during testing of samples.
- 42.3 All equipment shall be provided by the Contractor so as to be compatible with the testing requirements specified. The Contractor shall maintain the equipment in good working condition for the duration of the Contract.
- 42.4 The Contractor shall provide approved qualified personnel to operate and maintain the laboratory for the duration of the Contract. The number of staff and equipment available must at all times be sufficient to keep pace with the sampling and testing programme as required by the Engineer-in-Charge.
- 42.5 The Contractor shall fully service the site laboratory and shall supply everything necessary for its proper functioning, including all transport needed to move equipment and samples to and from sampling points on the site, etc.
- 42.6 The Contractor shall re-calibrate all measuring devices whenever so required by the Engineer-in-Charge and shall submit the results of such measurements without delay.

43. **Completion Certificate**

- 43.1 As soon as the Work is completed, the Contractor shall give notice of such completion to the Engineer-in-Charge and within eighty-four (84) days of receipt of such notice the Engineer-in-Charge shall inspect the Work and shall furnish the Contractor with a certificate of completion indicating (a) date of completion, (b) defects, if any, in the Work to be rectified by the Contractor and/or (c) items, if any, for which payment shall be made at reduced rates.

When separate periods of completion have been specified for items or groups of items, the Engineer-in-Charge shall issue separate completion certificates for such item or groups of items. No certificate of completion shall be issued nor shall the Work be considered to be complete till the Contractor shall have removed from the premises on which the Work has been executed all scaffolding, sheds and surplus materials (except such as are required for rectification of defects), and the like to the satisfaction of Engineer-in-Charge. If the Contractor shall fail to comply with any of the requirement of the conditions as aforesaid, on or before the date of completion of the Works, the Engineer-in-Charge may, at the expense of the Contractor fulfil such requirements and dispose of the scaffoldings, surplus materials and rubbish etc. as he thinks fit and recover the cost after giving due credit for the realised amount. The term 'completion' used herein means the physical completion of the Work and in no way means to connote the quality or time of performance of the Work.

- 43.2 If at any time before completion of the entire Work, items or groups of items for which separate periods of completion have been specified, have been completed, the Engineer-in-Charge can take possession of any part or parts of the same (any such part(s) being hereinafter in this Condition referred to as 'the relevant part') notwithstanding anything expressed or implied elsewhere in this Contract.

- 43.3 In case of such taking over of possession by Engineer-in Charge of the said item or group of items, the following shall govern:
- 43.3.1 Within twenty-eight (28) days of request by the Contractor, the Engineer-in-Charge shall issue completion certificate for the relevant part as in GCC Sub-Clause 43.1 above provided the Contractor fulfils his obligations under that Condition for the relevant part.
- 43.3.2 The Defects Liability Period in respect of such items and the relevant part shall be deemed to have commenced from the certified date of completion of such items or the relevant part as the case may be.
- 43.3.3 The Contractor may reduce the value insured under GCC Clause entitled 'Contractor's Liability and Insurance' to the extent of the value of the completed items or relevant part as estimated by the Engineer-in-Charge and notified for this purpose. This estimate shall be applicable for this purpose only and for no other.
- 43.3.4 For the purposes of ascertaining liquidated damages for delay under GCC Sub-Clause 29.6 in respect of any period during which the Works are not complete the relevant part will be deemed to form a separate item or group, with date of completion as given in the Contract or as extended under GCC Sub-Clause 29.2 and actual date of completion as certified by the Engineer-in-Charge under this Clause.

E. DEFECTS LIABILITY

44. Liability for Damage, Defects or Imperfections and Rectification thereof

- 44.1 If the Contractor or his workmen or employees shall injure or destroy any part of the building / structure in which they may be working or any building, road, fence etc. contiguous to the premises on which the Work or any part of it is being executed, or if any damage shall happen to the Work while in progress, the Contractor shall upon receipt of a notice in writing in that behalf make the same good at his own expense. If it shall appear to the Engineer-in-Charge or his representative at any time during construction or re-construction or prior to the expiration of the Defects Liability Period, that any work has been executed with unsound, imperfect or unskilled workmanship or that any materials or articles provided by the Contractor for execution of the Work are unsound or of a quality inferior to that contracted for, or otherwise not in accordance with the Contract, or that any defect, shrinkage or other faults have appeared in the Work arising out of defective or improper materials or workmanship, the Contractor shall, upon receipt of a notice in writing in that behalf from the Engineer-in-Charge, forthwith rectify or remove and re-construct the work so specified in whole or in part, as the case may require or as the case may be, and/or remove the materials or articles so specified and provide other proper and suitable materials or articles at his own expense, notwithstanding that the same may have been inadvertently passed, certified and paid for and in the event of his failing to do so within the period to be specified by the Engineer-in-Charge in his notice aforesaid, the Engineer-in-Charge may rectify or remove and re-execute the work and/or remove and replace with others, the materials or

articles complained of, as the case may be, by other means at the risk and expense of the Contractor.

45. Defects Liability Period

45.1 Unless otherwise specified in the SCC, the Contractor shall be responsible to make good and remedy at his own expense within such period as may be stipulated by the Engineer-in-Charge, any defect which may develop or may be noticed before the expiry of twelve (12) months from the certified date of completion.

45.1.1 On expiry of the period mentioned as defect liability period (GCC Clause 45), contractor's liability ceases except for latent defects. The contractor's liability for latent defect shall be limited to a period of five (5) years from the end of Defect liability period for the subject package. For the purpose of this clause the latent defects shall be the defects inherently lying within the material or arising out of design deficiency which do not manifest themselves during the defect liability period but may surface later.

F. RISK DISTRIBUTION

46. Employer's and Contractor's Risks and Insurance

46.1 The Employer carries the risks which this Contract states as Employer's risks, and the Contractor carries risks which this Contract states as Contractor's risks, under this clause.

46.2 Irrespective of the Employer's Risks or Contractor's Risks the Contractor shall execute the Works as per Contract and as directed by Engineer-in-Charge.

46.3 Employer's Risks

46.3.1 The 'Excepted Risks' are

- (1) In so far as they occur in the Union of India and directly affect the execution of the Works:
 - (a) war and hostilities (whether war be declared or not), invasion, act of foreign enemies.
 - (b) rebellion, revolution, insurrection or military or usurped power or civil war.
 - (c) riot, commotion or disorder, unless solely restricted to employees of the Contractor or of his sub-contractors and arising from the conduct of the Works;
 - (d) ionizing radiations, or contamination by radio activity from any nuclear fuel, or from any nuclear waste from the combustion of nuclear fuel, radioactive, toxic, explosive, or other hazardous properties of any explosive, nuclear assembly or nuclear component;

- (e) pressure waves caused by aircraft or other aerial devices traveling at sonic or supersonic speed;
- (f) any operation of the forces of nature, which is unforeseeable or against which an experienced contractor could not reasonably have been expected to have taken adequate precautions or

(2) a cause due to the design of the Works, other than the Contractor's design.

46.3.2 In the event of any loss or damage to the Works or any part thereof and/or to any materials or articles at the Site from out of any occurrence of Excepted Risks, the following provisions shall have effect:

- (a) The Contractor shall, as may be directed in writing by the Engineer-in-Charge, remove from the Site any debris and so much of the Works as shall have been damaged, take the same to the place identified by the Employer, at the Employer's cost.
- (b) The Contractor shall, as may be directed in writing by the Engineer-in-Charge, proceed to rectify, repair, reconstruct or replace the damaged articles, materials and the Works under and in accordance with the Conditions of the Contract, at the Employer's cost.

46.3.3 The Contractor shall not be entitled to payment under the above provisions in respect of so much loss or damage as has been occasioned by any failure on his part to perform his obligations under the Contract or not taking precautions to prevent loss or damage or minimize the amount of such loss or damage.

46.4 **Contractor's Risks**

46.4.1 All risks of loss of or damage to the physical property and of personal injury and death, which arise during and in consequence of the performance of the Contract, other than those covered under the Excepted Risks, will be the liability of the Contractor, except as otherwise provided in the Contract.

46.4.2 From commencement to completion of the Works, the Contractor shall take full responsibility for the care thereof and for taking precautions to prevent loss or damage and to minimize loss or damage to the greatest extent possible and shall be liable for any damage or loss that may happen to the Works or any part thereof from any cause whatsoever (save and except due to Excepted Risks) and shall at his own cost repair and make good the same so that at completion, the Works shall be in good order and condition and in conformity in every respect with the requirements of the Contract and instructions of the Engineer-in-Charge.

46.4.3 The Contractor shall indemnify and keep indemnified the Employer against all losses and claims for injuries or damage to any person or any property whatsoever which may arise out of or in consequence of the construction and maintenance of the Works and against all claims, demands, proceedings, damages, costs, charges and expenses whatsoever in respect of or in relation thereto; Provided always that

nothing herein contained shall be deemed to render the Contractor liable for or in respect of or to indemnify the Employer against any Compensation or damage caused by any occurrence of the Excepted Risks.

46.5 **Insurance**

46.5.1 Before commencing the execution of the Works, the Contractor shall, without in any way limiting his obligations and responsibilities under this clause, indemnify the Employer against any damage/ loss or injury which may occur to any property or to any person (including any employee of the Employer) by or arising out of carrying out of the Contract, except due to reasons of 'Excepted Risks'.

46.5.2 Towards this end, the Contractor shall arrange adequate insurance coverages, in the joint names of the Employer and the Contractor, from the date of commencement of the work to the end of the Defects Liability Period for at-least the following:

- (i) loss of or damage to the Works including Employer issued materials;
- (ii) loss of or damage to the Contractor's T&P;
- (iii) loss of or damage to the property other than Works including those of third parties;
- (iv) injury or death of personnel belonging to the Contractor, the Employer or any other party.
- (v) Worker's Compensation in accordance with the statutory requirements.

The Contractor shall be compulsorily required to take Contractor's All Risk Insurance Policy for all risks except those covered under the Excepted risks with minimal deductible for the re-execution value of the Works on completion and replacement cost of the Contractor's T&P/ Plant & Equipment/ Employer's free issue materials/ Third party Property, etc. as mentioned above. For this purpose the re-execution value of the Works on completion/ replacement cost of the Contractor's T&P/ Plant & Equipment/ Employer's free issue materials/ Third party Property shall include all such expenses, costs, taxes, duties, levies, royalties, etc. which in case of total loss, the Employer/ Contractor would be reasonably expected to incur to bring these to the same state as if no damage had occurred.

The Contractor and Employer shall mutually decide on the insured sum but in no case it shall be less than the 110% of the value of the Works on completion and Third party Property and 110% of the cost of Contractor's T&P/ Plant & Equipment/ Employer's free issue materials delivered to site including all taxes, duties, levies and royalties etc. and escalation during the period of re-execution for the Works and delivery of replacement Contractor's T&P/ Plant & Equipment/ Employer's free issue materials to site. However, in case of any conflict, the Employer's decision regarding the insured sum shall be final and binding on the Contractor. The Employer shall be named coinsured in the policy and it shall include the extended cover at-least for the following:

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- a) Third Party Liability
- b) Cross Liability
- c) Earthquake
- d) Cost of removal of debris
- e) Custom duty as applicable
- f) Escalation during re-execution
- g) Employer's surrounding property
- h) Storage risk at Fabricator's premises as applicable
- i) Cost of tools and tackles and material handling equipment used for construction purposes.
- j) War, SRCC & terrorism

- 46.5.3 Where Employer's building or a part thereof is rented by the Contractor, he shall insure the entire building if the building or any part thereof is used by him for the purpose of storing or using materials of combustible nature as to which the decision of the Engineer-in-Charge shall be final.
- 46.5.4 Contractor shall ensure that the insurance coverage of the above policies include any loss or damage to his Staff, Supervisors, Engineers and others who are not covered by Workmen Compensation Act. Alternatively, the Contractor will take suitable additional or separate insurance policies to cover the same.
- 46.5.5 All monies payable by the insurers under such policy or policies shall be first to be paid to the Employer who shall pay to the Contractor in instalments for the purpose of rebuilding or replacement or repair of the damaged Works, Contractor's T&P and/or materials destroyed or damaged as the case may be. Such payments of monies will be made in a manner that the payments are commensurate with the progress and cost of the repair, replacement and reconstruction, as may be and as determined by the Engineer-in-Charge in consultation with the Contractor.
- 46.5.6 Policies and certificate for insurance shall be delivered by the Contractor to the Project Engineer for the Project Engineer's approval before the date of commencement of the Works.
- 46.5.7 If the Contractor has a blanket insurance policy for all his works and the policy covers all or some of the items to be insured under this Clause, the said policy shall be assigned by the Contractor in favour of the Employer; provided however if any amount is payable under the policy by the insurers in respect of works other than the Work under this Contract, the same may be recovered by the Contractor directly from the insurers.
- 46.5.8 The aforesaid insurance policy/policies shall provide that they shall not be materially modified/ cancelled till the Engineer-in-Charge has agreed to such modification or cancellation in writing.
- 46.5.9 Upon grant of the time extension by the Engineer-in-Charge, it is understood that the Contractor's liability of indemnity will be extended suitably without any further action by the Employer and the Contractor shall promptly furnish documentary evidence to Engineer-in-Charge towards extension of insurance policies for the period of time extension.

- 46.5.10 The Contractor shall ensure that where applicable, his Sub-Contractor(s) shall take out and maintain in effect adequate insurance policies for their personnel and vehicles and for the part of the Works executed by them under the Contract, unless such Sub-Contractors are covered by the policies taken out by the Contractor.
- 46.5.11 If the Contractor and/or his Sub-Contractors (if any) shall fail to effect and keep in force the insurance coverage, referred to above or any other insurance which he/they may be required to effect under the terms of the Contract, then in any such case the Employer may, without being bound to, effect and keep in force any such insurance coverage and pay such premium or premiums, as may be necessary for that purpose from time to time and deduct the amount so paid by the Employer from any monies due or which may become due to the Contractor or recover the same as a debt due from the Contractor.
- 46.5.12 If the Contractor does not provide any of the policies and certificates required, the Employer, with due notice to the Contractor may effect the insurance which the Contractor should have provided and recover the premiums, the Employer has paid from payments otherwise due to the Contractor or, if no payment is due, the payment of the premiums shall be a debt due from the Contractor.

47. **Force Majeure**

47.1 **Definition of Force Majeure**

47.1.1 “Force Majeure” shall mean any event beyond the control of the Employer or of the Contractor, as the case may be, (but excluding ‘Excepted Risks’, which shall be dealt in accordance with GCC Clause entitled “Employer’s Risks”) and which they could not foresee or with a reasonable amount of diligence could not have foreseen and which substantially affect the performance of the Contract.

47.1.2 Notwithstanding the generality of the above, the following events shall be termed as Force Majeure events in respect of the Contract

- (i) terrorist acts,
- (ii) confiscation, nationalization, mobilization, commandeering or requisition by or under the order of any government or de jure or de facto authority or ruler or any other act of failure to act of any local state or national government authority,
- (iii) national/sectoral/illegal strike, sabotage, lockout, embargo, import restriction, port congestion, lack of usual means of public transportation and communication, industrial dispute, shipwreck, epidemics, quarantine and plague

47.2 **Notice of Force Majeure**

47.2.1 If either party is prevented, hindered or delayed from or in performing any of its obligations under the Contract by an event of Force Majeure, then it shall notify the other in writing of the occurrence of such event and the circumstances thereof

within fourteen (14) days after the occurrence of such event.

47.2.2 The party who has given such notice shall be excused from the performance or punctual performance is prevented, hindered or delayed.

47.2.3 Notwithstanding any other provision of the Clause, Force Majeure shall not apply to any obligations of the Employer to make payments to the Contractor herein.

47.3 Duty to Minimize Delay

47.3.1 The party or parties affected by the event of Force Majeure shall use reasonable efforts to mitigate the effect thereof upon its or their performance of the Contract and to fulfill its or their obligations under the Contract, but without prejudice to either party's right to terminate the Contract under GCC Sub-Clause 47.5.

47.4 Consequence of Force Majeure

47.4.1 If the Contractor is prevented from performing its obligations under the Contract by reason of Force Majeure of which notice has been given under Sub-Clause 47.2.1, and suffers delay by reason of such Force Majeure, the Contractor shall be entitled to an extension of time for any such delay, if the Completion is or will be delayed, in accordance with GCC Sub-Clause entitled "Extension of Time for Completion".

47.4.2 No delay or non-performance by either party hereto caused by the occurrence of any event of Force Majeure shall

- (a) constitute a default or breach of the Contract,
- (b) give rise to any claim for damages or additional cost or expense occasioned thereby

if and to the extent that such delay or non-performance is caused by the occurrence of an event of Force Majeure.

47.5 Termination for reasons due to extended Force Majeure

47.5.1 If the performance of the Contract is substantially prevented, hindered or delayed for a single period of more than seventy (70) days or an aggregate period of more than one hundred and forty (140) days or any such extended period as may be agreed to between the parties on account of one or more events of Force Majeure during the currency of the Contract, the parties will attempt to develop a mutually satisfactory solution, failing which either party may terminate the Contract by giving a notice to the other.

47.5.2 In the event of termination pursuant to GCC Sub-Clause 47.5.1, the rights and obligations of the Employer and the Contractor shall be as specified hereunder:

- (a) the Contractor shall be paid at contract rates for the work already executed by him
- (b) The Employer shall have an option to take over the Contractor's facilities/materials or any part thereof brought to site by the Contractor's facilities/materials or any part thereof brought to site by the Contractor, at

such rates as are determined reasonable by the Engineer-in-Charge.

- 47.5.3 In the event of any disagreement of the parties relating to matters at GCC 47.5.2, the dispute shall be settled in accordance with GCC Clause titled "Settlement of Disputes".

G. CHANGES IN CONTRACT ELEMENT

48. Changes in Constitution:

- 48.1 Where the Contractor is a partnership firm, prior approval in writing of the Employer shall be obtained before any change is made in the constitution of the firm. Where the Contractor is an individual or a Hindu Undivided Family business concern such approval as aforesaid shall likewise be obtained before the Contractor enters into any partnership firm which would have the right to carry out the Work hereby undertaken by the Contractor. If prior approval as aforesaid is not obtained, the Contract shall be deemed to have been assigned in contravention of GCC Sub-Clause 51.3 hereof and the same action may be taken and the same consequences shall ensue as provided for in the said GCC Sub-Clause 51.3.

49. Powers of Engineer-in-Charge for alterations/omissions/additions/ substitutions

- 49.1 The Engineer-in-Charge shall have power (i) to make alterations in, omissions from, additions to, or substitutions for the original specifications, drawings, designs and instructions that may appear to him to be necessary or advisable during the progress of the work, and (ii) to omit a part of the Works in case of non-availability of a portion of the Site or for any other reasons he may consider necessary and/or reasonable. Any such alterations, omissions, additions or substitutions shall be ordered by the Engineer-in-Charge as a deviation. The Contractor shall be bound to carry out the said deviation in accordance with instructions given to him in writing by the Engineer-in-Charge and such alterations, omissions, additions or substitutions shall form part of the Contract as if originally provided therein and shall be carried out by the Contractor on the same conditions in all respects on which he agreed to do the original Works, except as otherwise provided herein.

49.2 Permissible deviation limit for variations in Contract Items

- 49.2.1 In case of items of Work above ground surface, as it exists at the time of commencement of Work, quantities of which may change due to Site Conditions or any other reasons, the permissible limit of deviations over the original value of each item will be (+)20%.
- 49.2.2 In case of items of Work below ground surface, as it exists at the time of commencement of Work, quantities of which may change due to Site conditions or any other reasons, the permissible limit of deviations over the original value of each item will be (+)100% and (-30%).
- 49.2.3 For the purposes of GCC Sub-Clause 49.2.1 above, all the quantities of any item actually executed from 0-120% of the Contract quantity will be payable at Contract rates while the rates for the quantities above 120% will be subject to

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review/revision. Similarly, for the purposes of GCC Sub-clause 49.2.2 above, all the quantities of any item actually executed from 70% to +200% of the Contract quantity will be payable at Contract Rates while the rates for the entire quantity executed from 0% to 69% (if the total quantity executed is in this range) and the entire quantity executed above 200% will be subject to review/revision as provisions herein.

49.2.4 In case the Schedule of Quantities contains sub-items of Work under a Main Item, then the above permissible limits of deviation shall be applicable on the value of each such sub-item and not on the entire value of the Main Item.

49.2.5 The deviations up to the above permissible limits shall be carried out by the Contractor at the same rates and terms as per the Contract.

49.3 Methodology for Determination of Rates for variations of Contract Items beyond the permissible deviation limits

49.3.1 For Contract Items which exceed the limits over the original value of that item as mentioned in GCC Sub-Clause 49.2.1 & 49.2.2 above, the Contractor may, within fourteen (14) days of the date of receipt of the order to carry out the said work, inform the Engineer-in-Charge under advice to the Employer of the rate which he proposes to claim for such item(s) of Work on Market Rate(s) basis, supported by analysis of the rate claimed and the relevant documents to substantiate the same. The Engineer-in-Charge shall, within seventy (70) days thereafter, after giving due consideration to the rate(s) claimed by the Contractor, determine the rate(s), in consultation with the Contractor, on Market Rate(s) basis. In the event of disagreement between the Engineer-in-Charge and Contractor, even after the said seventy (70) days from the date of submission of claims of the rate(s) by the Contractor, the Engineer-in-Charge within a further period of twenty-one (21) days thereafter, shall fix the rate(s)/price(s) as are, in his opinion appropriate. The rate(s)/price(s) so fixed shall be notified to the Contractor and shall be final and binding.

49.3.2 If the Engineer-in-Charge fails to determine and notify the rate(s)/price(s) even after expiry of the said twenty-one (21) days, then the Contractor will be at liberty to refer the matter for resolution to the Employer within a further period of fourteen (14) days after the above said twenty-one (21) days. If the Employer does not determine and cause the Engineer-in-Charge to notify the rate(s)/price(s), then the matter would be determined in accordance with the provisions of GCC Clause entitled "Settlement of Disputes". However, in the meanwhile, the Engineer-in-Charge will pay for the items of Work executed beyond the permissible deviation limits, at 75% (seventy-five percent) of the rate(s)/price(s) claimed by the Contractor with satisfactory supporting documents or at Contract Rate, purely on adhoc and provisional basis subject to adjustment.

49.3.3 In the event of the Contractor failing to inform the Engineer-in-Charge, within the stipulated period of fourteen (14) days time, the rate(s) which he proposes to claim, supported by relevant documents to substantiate the same, the rate(s) for such item(s) shall then be determined by the Engineer-in-Charge in consultation with the Contractor (if he so desires) on the basis of Market Rate(s) within seventy (70)

days thereafter. The rate(s) /price(s) so determined shall be notified to the Contractor and shall be final and binding.

49.3.4 The provisions of GCC Sub-Clauses 49.3.1, 49.3.2 and 49.3.3 above shall only be applicable to such individual Contract Item(s)/ sub-item(s) of Work whose original value is equal to or in excess of 1% of the total Contract Value (as awarded).

49.3.5 For individual Contracts Item(s)/ sub-item(s) of Work whose original value as per Letter of Award is less than 1% of the total Contract Value (as awarded) for each such item, there shall be no limit on the extent of deviations over the original value of the Item and shall be paid on the contracted rate(s)/price(s).

49.3.6 Rates of Items of Work derived on the basis as detailed in GCC Sub-Clause 49.3.1 or 49.3.2, or notified under GCC Sub-Clause 49.3.3 shall not be eligible for price adjustment, provided the period of execution of such items of Work beyond the permissible deviation limit as per the schedule to be finalised in line with GCC Sub-Clause 49.5, is less than or equal to six (6) months.

49.3.7 Further, in case the period of execution of such items of Work is more than six (6) months, such items shall be eligible for price adjustment as per clause entitled 'Contract Price Adjustment'. The base date in such a situation shall be the date as specified by the Engineer-in-Charge while determination of the Market Rate.

49.4 **Methodology for Determination of Rates for Extra Items (Additional, Altered or Substituted Items) of Work**

49.4.1 Rates for Extra Items of Work (comprising of Additional, Altered or Substituted items of Work), shall be determined by the Engineer-in-Charge in the following order:

- (i) If the rate(s)/price(s) for extra items occurring in a particular schedule of quantities are available in other schedule of quantities forming part of the Contract, the lowest of such rate(s)/price(s) will be used, subject to the nature of work being comparable.
- (ii) If the rate(s) cannot be derived as per (i) above, then
 - (a) In case of contracts with only one Schedule of Quantities forming the part of the Contract, the rate(s)/price(s) for the extra item(s) shall be derived from the lowest of any similar item(s) in that Schedule.
 - (b) In case of contracts with two or more Schedules of Quantities forming a part of the contract, the rates for the Extra Item(s) will be derived from the nearest similar item appearing in the Schedule in which the extra item is to be executed failing which from any other Schedule in which nearest similar item is available, the rate so derived being the lowest of such derived from nearest similar items in those other Schedules and used.

49.4.2 If the rate for any additional, altered or substituted item of work cannot be determined

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in the manner specified in GCC Sub-Clause 49.4.1 (i) & (ii) above, the Contractor shall, within fourteen (14) days of the date of receipt of the order to carry out the said Work, inform the Engineer-in-Charge under advice to the Employer of the rate which he proposes to claim for such item(s) of Work on Market Rate(s) basis, supported by analysis of the rate claimed and relevant documents to substantiate the same. The Engineer-in-Charge shall, within seventy (70) days thereafter, after giving due consideration to the rate(s) claimed by the Contractor, determine the rate(s), in consultation with the Contractor, on Market Rate(s) basis. In the event of disagreement between the Engineer-in-Charge and Contractor, even after the said seventy (70) days from the date of submission of claims of the rate(s) by the Contractor, the Engineer-in-Charge within a further period of twenty-one (21) days thereafter shall fix the rate(s)/price(s) as are, in his opinion appropriate. The rate(s) /price(s) so fixed shall be notified to the Contractor and shall be final and binding.

49.4.3 If the Engineer-in-Charge fails to determine and notify the rate(s)/price(s) even after expiry of the said twenty-one (21) days, then the Contractor will be at liberty to refer the matter for resolution to the Employer within a further period of fourteen (14) days after the above said twenty-one (21) days. If the Employer does not determine and cause the Engineer-in-Charge to notify the rate(s)/price(s), then the matter would be determined in accordance with the provisions of GCC Clause entitled "Settlement of Disputes". However, in the meanwhile, the Engineer-in-Charge will pay for the extra items of Work, at 75% (seventy-five percent) of the rate(s)/price(s) claimed by the Contractor with supporting documents, purely on adhoc and provisional basis subject to adjustment.

49.4.4 In the event of the Contractor failing to inform the Engineer-in-Charge within the stipulated period of fourteen (14) days time the rate(s) which he proposes to claim, supported by relevant documents to substantiate the same, the rate(s) for such item(s) shall then be determined by the Engineer-in-Charge in consultation with the Contractor (if he so desires) on the basis of Market Rate(s) within seventy (70) days thereafter. The rate(s) /price(s) so determined shall be notified to the Contractor and shall be final and binding.

49.4.5 Rates of Extra Items of Work, derived from Schedule of Quantities as detailed in GCC Sub-Clause 49.4.1 (i) & (ii) above, shall be eligible for Price Adjustment as per GCC Clause entitled 'Contract Price Adjustment' including base date.

49.4.6 Rates for Extra Items of Work, derived on the basis as detailed in GCC Sub Clause 49.4.2 or 49.4.3, or notified under GCC Sub-Clause 49.4.4 above shall not be eligible for Price Adjustment, provided the period of execution of such Extra Items of Work beyond the permissible deviation limit as per the schedule to be finalised in line with GCC Sub-Clause 49.5, is less than or equal to six (6) months.

49.4.7 Further, in case the period of execution of such items of Work is more than six (6) months, such items shall be eligible for price adjustment as per clause entitled 'Contract Price Adjustment'. The base date in such a situation shall be the date as specified by the Engineer-in-Charge while determination of the market rate.

49.5 **Work Schedules for variation beyond deviation limits & Extra Items**

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49.5.1 The Engineer-in-Charge shall finalise a Work Schedule in consultation with the Contractor for items of Work beyond deviation limits and the Extra Items of Work to be executed and the date(s) specified in this agreed Work Schedule shall be considered as the date for working out the Price adjustment amount. The primary consideration by the Engineer-in-Charge while determining the time required for execution of the altered or substituted item(s) of Work, would be quantities of the altered or substituted and not the value of altered or substituted item(s) of Work. The Contractor shall not be eligible for Price Adjustment Payment for quantities of items executed beyond the schedule date(s), if execution of the items of the Work has been delayed for the reasons attributable to the Contractor.

49.5.2 However, the Contractor would be eligible for claim or liable for refund for Price Adjustment(s) for quantities of items of the Work executed beyond the schedule dates based on the value of indices as applicable to the scheduled dates of execution, for such delayed work provided that if the indices during the extended period are lower than the indices during scheduled period of execution, then lower indices shall be applicable.

49.6 **Provisional payments**

49.6.1 Pending approval of the Rates for Contract Item(s) of Work beyond the permissible deviation limits as well as for Extra Items (Additional, Altered or Substituted item) of Work, provisional payment at an interim rate (not exceeding 80% of the rate/price determined by the Engineer-in-Charge), shall be made to the Contractor in the interest of progress of Work, which shall be regularized after approval of Competent Authority.

50. **Suspension of Works**

50.1 The Contractor shall, on receipt of the order in writing of the Engineer-in-Charge, suspend the progress of the Works or any part thereof for such time and in such manner as the Engineer-in-Charge may consider necessary for any of the following reasons:

- (i) On account of any default on part of the Contractor; or
- (ii) for proper execution of the Works or part thereof for reasons other than the default on the part of the Contractor; or
- (iii) for safety of the Works or part thereof, for reasons other than those attributable to the Contractor.

50.2 The Contractor shall, during such suspension, properly protect and secure the Works to the extent necessary and carry out the instructions given in that behalf by the Engineer-in-Charge.

50.3 If the suspension is ordered for reasons (ii) & (iii) in Sub-Clause 50.1 above, in so far as it concerns suspension of part of the Works or whole of the balance, the Contractor shall be entitled to an extension of time equivalent to the period of suspension plus 25% thereof. The Contractor shall not be eligible for any other

compensation whatsoever for such suspension, except as otherwise provided herein under.

50.4 If the suspension is ordered for reasons (ii) & (iii) in Sub-Clause 50.1 above, as far as it concerns the entire balance of Works on the date of suspension and if such period of suspension cumulatively exceeds twenty-eight (28) days, then in addition to extension of time as in Sub-Clause 50.2 above, the Contractor shall be eligible for compensation, as the Employer may consider reasonable, in respect of salaries and/ or wages paid by the Contractor to his employees and labour at site, remaining idle during the cumulative period of suspension, adding to the total thereof, a reasonable percentage as determined appropriate by the Engineer-in-Charge, to cover indirect expenses and incidentals of the Contractor, provided the Contractor submits his claim supported by details to establish the reasonableness of his claim to the Engineer-in-Charge under advice to the Employer within fourteen (14) days of the expiry of the said twenty-eight (28) days period.

50.5 If for any reason other than for reasons of Contractor's default as per GCC Sub-Clause 50.1(i) above, if the Contract remains suspended for a continuous period exceeding ninety (90) days, then the Employer and the Contractor shall mutually discuss and agree for a suitable course of action regarding the recommencement/ reinstatement of the suspended work or alternatively treat the suspension as termination / abandonment of the Works by the Employer as per GCC Sub-Clause 51.1 herein. If out of above discussion it is determined that the Contract has to be treated as terminated under the provisions of GCC Sub-Clause 51.1, then the Contractor shall be eligible for compensation as envisaged in GCC Sub-Clause 51.1.1 herein.

51. Termination

51.1 Termination by the Employer

If at any time after award of Contract, the Employer shall decide to abandon or reduce the scope of the Works for any reason whatsoever and hence not require the whole or any part of the Works to be carried out by the Contractor, the Engineer-in-Charge shall give notice in writing to that effect to the Contractor and the Contractor except as herein under provided, shall have no claim to any payment of compensation or otherwise whatsoever, on account of any profit or advantage which he might have derived from the execution of the Works in full but which he did not derive in consequence of the said termination of the whole or part of the Works.

51.1.1 The Contractor shall be paid at Contract rates full amount for works executed at Site and, in addition, a reasonable amount as certified by the Engineer-in-Charge for the items hereunder mentioned which could not be utilised on the Work to the full extent because of the said termination:

- (a) Any cost incurred on preliminary site work, e.g. access roads, labour huts, staff quarters and site offices; storage accommodation and water storage tanks, etc.

- (b) (i) The Employer shall have the option to take over Contractor's facilities/ materials or any part thereof either brought to Site or of which the Contractor is legally bound to accept delivery from suppliers (for incorporation in or incidental to the Work), provided, however, the Employer shall be bound to take over the materials or such portions thereof as the Contractor does not desire to retain. For materials taken over or to be taken over by the Employer, cost of such materials shall, however, take into account purchase price, cost of transportation and deterioration or damage which may have been caused to materials whilst in the custody of the Contractor.
- (ii) For Contractor's materials not retained by the Employer, reasonable cost of transporting such materials from Site to Contractor's permanent stores or to his other Works, whichever is less. If materials are not transported to either of the said places, no cost of transportation shall be payable.
- (c) If any materials issued by the Employer are rendered surplus, the same except normal wastage shall be returned by the Contractor to the Employer at rates not exceeding those at which these were originally issued less allowance for any deterioration or damage which may have been caused whilst the materials were in the custody of the Contractor. In addition, cost of transporting such materials from Site to the Employer's stores, if so required by the Employer.
- (d) Reasonable compensation for transfer of Contractor's T&P from Site to Contractor's permanent stores or to his other Works, whichever is less. If T&P are not transported to either of the said places, no cost of transportation shall be payable.

51.1.2 The Contractor shall, if required by the Engineer-in-Charge furnish to him wage books, time sheets and other relevant documents as may be reasonably necessary to enable him to certify the reasonableness of the amount payable under this Clause.

51.2 **Termination on Contractor's Death**

51.2.1 If the Contractor is an individual or a proprietary concern and the individual or the proprietor dies and if the Contractor is a partnership concern and one of the partners dies, then unless the Employer is satisfied that the legal representatives of the individual Contractor or of the proprietor of the proprietary concern and in the case of partnership, the surviving partners, are capable of carrying out and completing the Contract, the Employer shall be entitled to cancel the Contract as to its incomplete part without the Employer being liable in any way to payment of any compensation to the estate of the deceased Contractor and/or to the surviving partners of the Contractor's firm on account of the cancellation of the Contract. The decision of the Employer that the legal representatives of the deceased Contractor or the surviving partners of the Contractor's firm cannot carry out and complete the Contract shall be final and binding on the parties. In the event of such cancellation the Employer shall not hold the estate of the deceased Contractor and/or the

surviving partners of the Contractor's firm liable for damages for not completing the Contract.

51.3 Termination for Contractor's Default

51.3.1 If the Contractor:

- (a) at any time makes default in proceeding with the Works with due diligence and continues to do so after a notice of seven (7) days in writing from the Engineer-in-Charge; or
- (b) commits default in complying with any of the terms and conditions of Contract and does not remedy it or take effective steps to remedy it within seven (7) days after a notice in writing is given to him in that behalf by the Engineer-in-Charge; or
- (c) fails to complete the Works or items of Work with individual dates of completion, on or before the date(s) of completion, and does not complete them within the period specified in a notice given in writing in that behalf by the Engineer-in-Charge; or
- (d) shall offer, or give or agree to give to any person in Employer's service or to any other person on his behalf any gift or consideration of any kind as an inducement or reward for doing or forbearing to do or having done or forborne to do any act in relation to the obtaining or execution of this or any other Contract for the Employer; or
- (e) shall enter into a contract with the Employer in connection with which commission has been paid or agreed to be paid by him or to his knowledge, unless the particulars of any such commission and the terms of payment thereof have been previously disclosed in writing to the Employer/ Engineer-in-Charge; or
- (f) shall obtain a Contract with the Employer as a result of ring bidding or other non-bona fide methods of competitive bidding; or
- (g) being an individual, or if a firm, any partner thereof, shall at any time be adjudged insolvent or have a receiving order or order for administration of his estate made against him or shall take any proceedings for liquidation or composition (other than a voluntary liquidation for the purpose of amalgamation or reconstruction) under any Insolvency Act for the time being in force or make any conveyance or assignment of his affective or composition or arrangement for the benefit of his creditors or purport so to do, or if any application be made under any Insolvency Act for the time being in force, for the sequestration of his estate or if a trust deed be executed by him for benefit of his creditors; or
- (h) being a company, shall pass a resolution or the Court shall make an order for the liquidation of its affairs, or a receiver or manager on behalf of the debenture holders shall be appointed or circumstances shall arise which entitle the Court or debenture holders to appoint a receiver or manager; or

- (i) assigns, transfers, sublets (engagement of labour on a piece-work basis or of labour with materials not to be incorporated in the work, shall not be deemed to be subletting) attempts to assign, transfer or sublet the entire Works or any portion thereof without the prior written approval of the Employer; the Employer may, without prejudice to any other right to remedy which shall have accrued or shall accrue thereafter to the Employer by written notice, cancel the Contract as a whole or only such items of work in default, from the Contract.

(j) **“Withholding/ Banning**

The Employer has in place a Policy for withholding and Banning of Business Dealings as enclosed at ANNEXURE-I to Special Conditions of Contract (SCC) of the Bidding documents. Business dealings may be withheld or banned with the Contractor on account of any Default by the Contractor under GCC Clause 51.3.1 or any of the grounds as detailed in the said Banning Policy.”

Encl.: Annexure-I to SCC

51.3.2 The Employer shall on such cancellation have rights to:

- (a) take possession of the Works and any materials, construction plant, implements, stores, etc., thereon; and/or
- (b) carry out the incomplete Work by any means at the risk and cost of the Contractor.

51.3.3 On cancellation of the Contract in full or in part, the Employer shall determine what amount, if any, is recoverable from the Contractor for completion of Works or part of the Works or in case the Works or part of the Works is not completed, the loss or damage suffered by the Employer. In determining the amount, credit shall be given to the Contractor for the value of the work executed by him up to the time of cancellation, the value of Contractor’s material taken over as well as incorporated in the work, and use of tools and plants belonging to the Contractor.

51.3.4 Any excess expenditure incurred or to be incurred by the Employer in completing the Works or part of the Works or the excess loss or damages suffered or may be suffered by the Employer as aforesaid after allowing such credit shall be recovered from any money due to the Contractor on any account, and if such money is not sufficient the Contractor shall be called upon in writing to pay the same within twenty-eight (28) days.

51.3.5 If the Contractor shall fail to pay the required sum within the aforesaid period of twenty-eight (28) days, the Engineer-in-Charge shall have the right to sell any or all of the Contractor’s unused materials, construction plant, implements, temporary buildings etc. and apply the proceeds of sale thereof towards the satisfaction of any sums due from the Contractor under the Contract and if thereafter there be any balance outstanding from the Contractor, it shall be recovered from him.

51.3.6 Any sums in excess of the amounts due to the Employer and unsold materials, construction plant etc., shall be returned to the Contractor, provided always that if cost or anticipated cost of completion by the Employer of the Works or part of the

Works is less than the amount which the Contractor would have been paid had he completed the Works or part of the Works, such benefit shall not accrue to the Contractor.

52. **Contractor Performance and Feedback System**

The Employer has in place an established 'Contractor Performance and Feedback System' against which the Contractor's performance during the execution of Contract shall be evaluated on a continuous basis at regular intervals on the following seven parameters:

- Engineering & Quality Assurance Capability
- Finance • Supply
- Construction/ Installation
- Field Quality
- Safety
- Claims & Disputes

The score-based feedback formats based on which Contractor's performance shall be evaluated is enclosed at Annexure-A.

In case the performance of the Contractor is found unsatisfactory, the Contractor shall be considered ineligible for participating in future tenders for three years.

On completion of the above ineligibility period, the Contractor would be required to submit a request to NTPC for participating in future tenders specifying the measures taken to improve their performance. The Contractor may also request for early revocation of suspension after completion of at least two (2) year of the suspension period. On receipt of such request, the performance of the Contractor shall be assessed/evaluated by NTPC and if the performance is found to be satisfactory, the Contractor shall be considered eligible for participation in future tenders.

53. **Withholding / Banning**

The Employer has in place a Policy for withholding and banning of Business Dealings as enclosed at **Appendix-I** to this section. Business dealings may be withheld or banned with the Contractor on account of any Default by the Contractor under Clause 51.3.1 or any of the grounds as detailed in the said Banning Policy.

54. **CONTRACTOR'S LABOUR INFORMATION MANAGEMENT SYSTEM (CLIMS):**

- (a) The Contractor has to necessarily get itself registered in the Contractor's Labour Information Management System (CLIMS), which will be installed by the Employer.
- (b) The entry and exit of all contract labour to the plant premises will be through Gate Access Control System of above 'Contractor's Labour Information Management System'.
- (c) It will be the responsibility of the Contractor to ensure timely exit of all labours from the plant premises after completion of job of that day.

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- (d) The contractor has to abide with all the statutory compliance applicable to its workers and employees and update the details of the same in the above System.

Contractor Safety Manual



JHABUA POWER LIMITED
Barela, Seoni (MP)

MISSION : Zero Harm
No Harm to any JPL Stakeholder

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1.0 INTRODUCTION/PURPOSE

- ❖ Jhabua Power Limited (JPL) is going to implement this Contractor Safety Manual to allow the contracting Organization to conduct the business at JPL as safely as possible. The manual is an agreement between JPL & the contracting agency to achieve the mission of Zero Harm.
- ❖ We **INSIST** that the contractor agency must familiarize his people with the contents of this manual and abide by all Rules and Guidelines mentioned there in the manual.
- ❖ Contractor shall be solely responsible for the safety of his employees, sub-contractors and agents during execution of the work

2.0 SCOPE

The JPL Contractor Safety Manual applies to all the Contractors providing services for Jhabua Power Limited.

3.0 DEFINITIONS

- JPL: Jhabua Power Limited
- The Factories Act: The Factories Act, 1948
- Contractor: Organization who is contracted by JPL to perform work or services for JPL at JPL premises.
- Subcontractors: Organization that is contracted by the Contractor to perform work or services for JPL at JPL premises.
- Contractor Employee: Any employee or service provider of a Contractor or Subcontractor.
- Side Incharge (Contractor Lead): The person designated by a Contractor and is responsible for Safety of the Contractor Employees.
- JPL Contact (EIC, Engineer in charge): The JPL employee who is responsible for JPL's contract with the Contractor. This responsibility may be designated, but the contact will be clearly identified prior to start of the work.

4.0 GENERAL RESPONSIBILITIES:

- JPL intends to hire only those Contractors who can provide services in a safe and healthy manner. By accepting a contract, the Contractor commits that he has the required skills, experience, expertise and commitment to perform work in a safe and healthy manner, and will allow only those Contractor Employees who are properly trained and supervised to work at JPL premises.
- If a contract between JPL and a Contractor allows for the Contractor to subcontract, the Subcontractor must have all the skills, experience, expertise and commitment to perform work in a safe and healthy manner, and that the Contractor will allow only those Subcontractor Employees who are properly trained and supervised to work at JPL premises. The Contractor is also responsible for providing this Contractor Safety Manual to the Subcontractor before work on JPL premises begins.
- A Contractor is responsible to keep JPL informed about its activities, and the activities of its Subcontractors. This is to be accomplished by the Side Incharge (Contractor Lead) with the JPL Contact.
- JPL is not responsible for safety and health policies or practices of any Contractor or Subcontractor. This Manual is to provide guidance to Contractors and Subcontractors on how they can satisfy their own

responsibilities in this regard.

5.0 SAFETY REQUIREMENTS:

- Contractor shall have sufficient knowledge, experience and understanding of thermal power plant work practices, safety & health hazards and other regulatory requirements pertaining to the work to be performed.
- Contractor shall perform the work using qualified workers who are adequately trained in the requirements of their particular job and skilled in the work assigned to them.
- Contractors shall provide proof of worker credentials (training, Qualification certificates, license etc.) on request/joining.
- Contractor shall comply with the requirement of Factory Act 1948, M P Factory Rules 1962 and other central & State laws, rules, regulation & time to time released orders of governing authority.
- Contractor shall comply with all company posted requirements, information provided by the company on site specific hazards and emergency response plans.
- Contractor shall review this manual with his employees, sub-contractors and consultants.
- Contractor shall have dedicated safety representative at the work site all the times. Contractor shall provide the qualifications of the proposed safety representative to company for review and approval.
- Contractor dedicated qualified safety representative shall perform inspection of operations, facilities and equipment's used in the performance of the work and participate in joint inspections, audits with company on request. Contractor shall immediately address any unsafe conditions, equipment's or action identified during inspection.
- Contractors shall ensure workers comprehend job specific safety related information including individuals in English, Hindi, Punjabi or any language which is easily understandable.
- Regularly Scheduled Safety Meetings: Contractor shall conduct regularly scheduled safety meetings. Attendance shall be required by all workers. Contractor shall keep a written record of the meetings that includes date, location, names or signatures of attendees, and topics covered. Contractor shall inform workers of factual circumstances resulting in incidents and near misses and discuss how to correct and prevent such situations from recurring.
- Daily Toolbox talk Meetings: Contractor shall conduct and document a daily morning safety meeting with all applicable workers to discuss Work activities, address any safety and health concerns for the Work to be performed, review any near miss incidents and how they could have been avoided, and prepare or review the appropriate Job Safety Analysis. Contractor shall provide such documentation to JPL upon request.

6.0 WORKPLACE REQUIREMENTS:

- Professional Conduct - Contractor shall conduct itself in a professional manner at all times while on Company Property. Horseplay, practical jokes, any type of harassment, abusive or objectionable language, or other inappropriate behavior on the job shall not be tolerated.
- Consequences for Non-Compliance - Working safely is a condition of employment at JPL. Any Contractor violating these rules and/or procedures will be required to permanently leave JPL premises.
- Weapons - All firearms, knives and other weapons are strictly forbidden at all JPL premises.

- Acts and threats of violence (physical or verbal) are strictly prohibited.
- Contractor shall inform its employees, suppliers, and subcontractors before entering Company Property that Company and its authorized representatives can search the person, vehicle, and other property of individuals while entering or departing from Company Property.
- The possession or use of narcotics, drugs, or intoxicating beverages of any kind is prohibited on JPL premises. Workman of Contractor shall immediately remove from the Workplace, if found in drug and alcohol testing violation.
- Contractor shall not bring unauthorized individuals (e.g., friends, relatives, or observers) onto Company premises.

7.0 SITE ACCESS

- Each Contractor Employee must carry his ID card when entering or providing services at JPL.

8.0 SAFETY PROGRAM

- The Contractor will have a safety program that outlines the requirements for performance of Contractor Employees specific to their activities. The Contractor is responsible for ensuring that this safety program meets the requirements of law, including but not limited to compliance with applicable Factories Act and other legal requirements.
- The Contractor will ensure that Contractor Employees have been trained prior to performing any activity at JPL
- The Contractor will ensure that its employees know the requirements outlined in this manual prior to beginning any work activity.
- The Contractor will communicate specific hazards found at JPL that may affect the safe work of Contractor Employees (e.g., working with chemicals, working in confined places, Electrical Hazards).
- The Contractor will be responsible for the direct supervision of Contractor Employees.

9.0 EMERGENCY RESPONSE AND EQUIPMENT

- Access to exits and to any emergency equipment (e.g., safety showers, eyewash fountains, firefighting equipment) must be kept clear at all times.
- The Contractor is responsible for communicating emergency procedures to Contract Workers. At all JPL facilities, immediate evacuation is required when audible alarm is sounded and/or an announcement is made to evacuate.
- Contractor Employees must leave by the closest/safest exit, as quickly as possible. After exiting the building, Contractor Employees must assemble in marked Safe Assembly Point and report immediately to the Side Incharge (Contractor Lead).
- The Side Incharge (Contractor Lead) is responsible for informing their JPL contact that the evacuation of all Contractor Employees was successful.

10.0 SPILLS AND INCIDENTAL RELEASES OF HAZARDOUS MATERIALS

- Contractor Employees must report any spill of chemicals or hazardous materials to their JPL contact immediately.
- A Contractor bringing any chemical or hazardous material onto JPL premises is required to provide their JPL contact with an MSDS for each substance and to use only approved, labeled containers for these substances.
- Provisions for spill prevention, response, and disposal of wastes generated from any potential clean-up activities from Contractor chemicals / hazardous materials must be discussed with the JPL contact before starting work with these substances.

11.0 PERSONAL PROTECTIVE EQUIPMENT (PPE)

- **General:** "PPE Hazard Assessment" means the process of identifying, selecting, and documenting appropriate personal protective equipment (PPE) for workplace hazards.
- Contractor shall provide, at its own expense, and enforce the use of all appropriate job specific PPE's and any posted Company requirements.
- Contractor shall ensure that all workers are trained in the proper use of applicable PPE's prior to performing Work.
- Approved hard hats, hard-toed boots or shoes, and safety glasses shall be worn on all Company worksites.
- Loose or floppy clothing, neck chains, loose jewelry, or loose long hair is prohibited.
- Rings shall be removed when working in areas where they could catch on moving objects or sharp protrusions or come into contact with electrical circuits.
- Clothing, including gloves, shall not be cleaned by blowing with compressed air, blowing with compressed gases, or washing in a flammable liquid

TO HAVE UNIFORMITY IN THE USAGE OF PPE'S, APPROVED MODELS AND MAKE OF PPE'S AS MENTIONED BELOW ARE ACCEPTABLE AT JPL WORKPLACE.

LIST OF PPES & SAFETY EQUIPMENT'S (APPROVED MAKE/MODEL)			
S.No.	Item	Make/Model	Photograph
1	Industrial Safety Helmet	Joseph leslico / Karam/ 3M/Savior make with ratchet. IS 2928 & EN 397 approved Qty: White for Engineer / Supervisor, Yellow for Workman, Red for Electrician, & Green for Safety.	
2	Safety Shoes	LIBERTY make "WARRIOR" brand / Allen Cooper / Karam / Udyogi/ Bata low-cut 2mm thick leather safety shoes with IMPORTED Esjot alloy toecaps & Double density directly injected PU black & Grey sole/ Safety shoe can allow ISI mark. As IS:15298	
3	Ear Plug	Honeywell / Karam/ 3M™ E-A-R™ UltraFit™ cloth/cotton Corded Earplugs, in poly bags/Carrying Case 2. Reusable, Triple flange design. NRR 25 dB	

		Karam EP 02 / 3M 32,dB	
4	Ear muff	Leslico / Karam /3M Hard hat mounted 1450 ear muff- NRR 23 dB	
		Helmet attachable ear muff EP 23, NRR 27 dB, EN 352- 1; 2002	
5	safety spectacles (shaded/clear)	Karam / 3M™ Virtua™ / UVEX Protective Eyewear, 11326-00000-100 Clear Temples Clear Hard Coat Lens 100 ea/case 3M™ Virtua™ Protective Eyewear, 11327-00000-20 Gray Hard Coat Lens, Gray Temple 20 ea/case Karam/3M™ Virtua™ / UVEX Protective Eyewear, 11329-00000-20 Clear Anti-Fog Lens, Clear Temple 20 ea/case Kara / 3M™ Virtua™ / UVEX Protective Eyewear AP, 11815-00000-20 Gray Hard Coat Lenses 20 ea/cs' & 3M™ Virtua™ Protective Eyewear AP, 11819-00000-20, Clear Hard Coat Lenses 20 ea/cs'	 
6	Goggles (for chemical handling)	Karam make / UVEX/ 3M etc. ES 009 clear, Confirms to EN 166:2001 Karam / UVEX / 3M 1621 safety goggles for splashes (can be worn over prescription lens). ANSI Z87.1-2003	
7	Respirators (dust mask)	3M / Leslico / Venus 8210 dust respirator, N 95 , NIOSH approved, ISI Approved.	
8	Welding Respirators	3M / Leslico / Venus welding respirator 8512, NIOSH approved	
9	Half face mask	3M / Venus half face piece reusable respirator 6200 series (medium size)(to be used with cartridges),NIOSH approved	

10	Chemical cartridge With attachment and retainer	3M / Venus organic vapour/acid gas cartridge 6003,NIOSH approved 3M N95 particulate filter 5N 11, NIOSH approved	
11	Reflective jackets (for own staff)	Reflectosafe With Reflective tape : 3M / Tango / Udyogi , Reflective Tape: 5 cm wide, Total length – 232 cm. Colour – Fluorescent Green	
12	Reflective jackets (for workers)	Reflective Tape - Micro prismatic reflective tapes 5 cm wide, total length - 260 cm. Front Opening Type Colour - Fluorescent Orange	
13	Cotton gloves	Cotton PVC Dotted Gloves make Midas / LESLICO / 3M / Honeywell For general handling/maintenance in good quality. Type: 7 Gauge Size: 25 cms Weight: 60 gms/pair Dott Colour: Blue	
14	Gloves (acid/alkali, cut resistant, leather)	Acid/alkali gloves, Leather hand gloves, Nitrile gloves, cut resistant Kevlar gloves, dotted gloves, welding gloves make Midas/ LESLICO / 3M / Honeywell / Polysol	
15	Chemical protective clothing	Chemical protective clothing by Udyogi plastics / DuPont Tychem / Microgard / Microchem	
16	Electrical safety gloves	Sperian electrosoft (marketed by Suresafety) or Honeywell , Beige natural latex insulating glove.CE certified, EN 60903. Must be used with a leather underglove. Class 4:- Category AZC.Thickness: 3.4 mm. Handling of high voltage to 36 000 volts. Class 3:- Category RC.Thickness: 2.9 mm. High voltage to 26 500 volts Class 2:- Category RC.Thickness : 2.3 mm. High voltage to 17 000 volts Class 1:- Category RC.Thickness 1.5 mm maximal voltage of 7500 volts Class 0:- Category RC. Thickness 1 mm. up to 1,000 volts	

		Honeywell / 'Saviour' Electrex Model No.– ELECTREX-33, ERDA Approved-Tested as per IS: 4770, 1991 (marketed by Suresafety), Provides protection from 33 KV Test voltage Electrical operations	
		Honeywell / 'Saviour' Electrex Model No.– ELECTREX-11, ERDA Approved-Tested as per IS: 4770, 1991 (marketed by Suresafety), Provides protection from 11 KV Test voltage Electrical operations	
17	Full Body Safety harness	Karam / Unicare / Udyogi brand full body harness model: Rhino PN 23 with PN 351 Double Lanyard Scaffolding hook and energy absorber. Conforms to EN 361: 2002, CE approved.	
18	Retractable wire rope fall arrester	Karam / Unicare/ Udyogi brand, PCGS 15, PCGS 20, PCGS 25. EN 360	
19	Fall arrester with energy absorber	Karam / Unicare / Udyogi PN 2000 B. EN 353	
20	Anchorage webbing slings	Karam / Unicare / Udyogi brand Concrete anchor strap PN 805 and PN 806 (lengths 1.0m, 1.5 m, 2.0 m). EN 795	
21	Anchorage SS wire rope	Anchorage SS wire rope Karam PN 814. EN 795 or Unicare / Udyogi	
22	Safety net	Karam / Garware/ Udyogi / Safetech make Safety net made from Polypropylene ropes. Mesh size:- 25 mm & 100 mm, Size:- 10 m X 5 m, mesh rope:- 2mm/4mm double cord, with overlay net	
23	Temporary horizontal lifeline	Karam / Unicare / Udyogi PN 3000, EN 795 Class B, made up of 30 mm polyester webbing and ratchet tensioner. Both ends fitted with auto locking steel karabiners.	
24	Vertical lifeline	Karam / Unicare / Udyogi, 3 strand polyamide twisted rope of dia 14 mm, one end loop and other end stop knot. 10 m to 200 m (PN 910 to PN 9200)	

25	Work positioning lanyard	Karam / Unicare / Udyogi make work positioning lanyard with ring type adjuster PN 241 . Made of 14 mm dia polyamide 3 strand twisted rope. Steel karabiner PN 112 at both ends. Manual ring type adjuster. EN 358	
26	Fire blankets	Udyogi/ Karam etc. make Fire blanket compact, wall mounted, easy to use made of asbestos free Material in size : 1.2mtr X 1.8mtr	
27	Gas welder's glass	Karam ES 003 / Honeywell/ Unicare fitted with IR 5 lens. EN 166	 ES 003 (IR 5)
28	Arc welder's glass	Karam / Honeywell / Unicare ES 004 fitted with IR 11 lens. EN 166	 ES 004
29	Helmet attachable welding shield	Karam / Honeywell make ES 71. It fits standard helmet with 30 mm slot. Confirms to EN 175 and ANSI Z 87.1. (To be fitted with IR 5 / IR 11 lens, sandwiched between two polycarbonate lens, confirming to EN 166 and ANSI Z 87.1)	 Ref. ES 71
30	Helmet attachable grinding shield	Karam ES 51 (02)/ Honeywell or Joseph Leslico/ equivalent. CE certified	 Ref. ES 51
31	Eye Wash Bottle	Unicare make UEWB 12 / Udyogi EW/ equivalent – 500 ML	
32	Barricading tape	Made up of virgin quality pvc material tubing pattern. Roll red & white colour with "DANGER/STOP & CAUTION/WORK IN PROGRESS" letters in Hindi &English Size : 3" Make : SAFE-T-PLUS / PRIMA or equivalent	
33	Life buoy	Unicare or equivalent The Life Buoy is manufactured in durable high visibility orange synthetic material and requires no maintenance. Moisture proof and non-inflammable. The H-Glow reflective material fixed in the indentations identifies victims in distress. Approved by the Mercantile Marine Department to SOLAS spec.	

34	Life jacket	Unicare or equivalent •MMD approved to Solas Specifications •With Solas Reflective Tape, Signaling Whistle and Light Quick turning time and high free board in water •Buoyancy : 17.5 Kg •Light that is automatically activated when in contact with sea water/ petroleum products. •Size : 80 X 37 X 10cms. Approved by the Mercantile Marine Department to SOLAS spec.	
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12.0 ELECTRICAL SAFETY

- Qualified Person – A designated worker who has the skills and knowledge related to operation of the electrical equipment and installations to be employed who should have received training to recognize and avoid the electrical hazards involved. Usually, this is a licensed electrician or someone with certain equivalent experience and training. A person can be “qualified” to work with certain equipment and methods and still be “unqualified” for other work.
- Contractor shall perform all electrical Work in accordance with the current editions of applicable central, state and local laws, rules, regulations, and standards.
- Installation of electrical systems or modifications to electrical systems shall be done under the supervision or direction of a licensed electrician.
- Contractor shall ensure that workers near overhead power lines know the voltage of the line and the safe approach distance.
- Contractor shall be aware of, and take precautions to prevent, the build-up of static electricity in areas with a potential Hazardous Atmosphere.
- Only qualified Person shall discharge all stored electrical energy and shall verify that equipment is de- energized and proper Lockout/ Tagout (LO/TO) procedures have been implemented prior to beginning electrical Work.
- All power lines shall be considered energized unless proper measures have been taken to de-energize.

13.0 LOCKOUT/TAGOUT (LOTO)

- Contractor shall ensure compliance with all requirements of company LOTO procedure.
- Contractor shall ensure that its workers are adequately trained in LOTO and applicable energy control procedures.
- LOTO energy control procedures shall be followed prior to work on any equipment or process where stored energy or the unexpected energizing of equipment could cause injury to a worker. Potential energy sources include electrical, mechanical, pneumatic, hydraulic, thermal, chemical, natural gas, and all forms of potential and stored energy.
- Repairs, maintenance, or alterations shall not be made on equipment in operation. All equipment shall be shut down and a LOTO device placed in a manner that the equipment cannot be accidentally started.

- Contractor shall ensure that a briefing is conducted with all workers affected by a LOTO operation before each shift, and more frequently if warranted by personnel changes or changes in the scope of Work. The briefing should include the following items:
 - i. The specific equipment or process involved, along with any related equipment;
 - ii. The estimated length of time required to complete the task;
 - iii. The hazards involved in performing repairs or maintenance, including the potential hazards to workers if the equipment or process is prematurely energized; and
 - iv. A review of the site-specific energy control procedure.
- To ensure the machine or equipment has been properly locked out of service prior to starting any Work, a Qualified Person shall attempt to turn on the power source to verify that the machine or equipment does not become energized.

14.0 HOT WORK

- "Hot work" means riveting, welding, flame cutting or other fire or spark-producing operation. No hot work is to be performed without first obtaining an JPL Hot Work permit, except in designated locations including Workshop and facilities Maintenance areas.
- Your JPL contact will help you identify the need for and to complete required permit requirements.

15.0 COMPRESSED GAS SAFETY

- Compressed gas cylinders must be secured in an upright position and kept away from sources of heat or flame at all times.
- All compressed gas cylinders must be legibly marked with either their chemical or trade name.
- All compressed gas cylinders not in use must have their top caps securely tightened.
- Where different gases are stored, they must be grouped by types. Groupings shall separate the flammable gases from the oxidizing gases.
- All oxygen cylinders must be stored not less than 20 feet from fuel gas cylinders or other flammable gasses.
- Compressed gas cylinders may not be dropped, dragged, rolled on their side or struck violently.
- A compressed gas cylinder may only be lifted by cradles or enclosed platforms when using a crane or hoisting device.
- A frozen or ice-clogged valve shall be thawed either by warm air or use of warm water and dried before using. Boiling water or a flame shall not be used. Force shall not be applied to a valve or cap to loosen a cylinder frozen in place.

- A cylinder shall not be placed where it will become a part of the electrical circuit by accidental grounding or where it may be burned by electric welding arc. A cylinder shall not be placed so that hot slag or flame will reach it or it shall be protected by a fire resistant shield.
- Flammable gas cylinders including LPG cylinders may not be stored within 25 feet of an emergency exit.
- Flashback arrestors/preventers are required on all cylinders and torches.

16.0 CONFINED SPACES

- Confined space entry requires an JPL Confined Space Entry Permit. The Contractor is responsible for compliance with the standards for any entry by a Contractor Employee.
- Confined space entry training is also a prerequisite for any entrants or attendants as part of this program.
- Entry equipment must be provided by the Contractor and calibration information must be available upon request.
- Emergency rescue procedures, equipment and personnel are the responsibility of the Contractor.
- Entry / Exit attendance format to be filled.

17.0 LADDERS

PORTABLE LADDERS:

- Contractor shall ensure that all workers have been trained in the proper use, placement, care and maximum load carrying capacities of the ladders used.
- Contractor shall inspect all ladders before use. Any damaged or unsafe ladders shall be tagged and taken out of service.
- Portable ladders shall be set at the correct angle. The distance from the foot of the ladder to the structure should be equal to 1/4 the length of the ladder.
- Workers shall keep both hands free for climbing, descending, and performing Work on a ladder. Carrying hand tools or equipment while climbing on a ladder is prohibited unless secured in a pocket or on a belt. Articles that are too large to be carried in a pocket or on a belt shall be lifted and lowered by a hand line.
- Workers shall not rush, but shall take one step at a time and face the ladder while climbing and descending.
- Only one person at a time shall be on the ladder.
- Portable ladders shall have anti-slip safety feet and be secured at the top before work begins in order to prevent the ladder from shifting. A second person shall hold the ladder until the climber can secure it at the top.
- Ladders shall extend at least 3 feet above the point of support when gaining access to a roof or other area.
- Contractor shall use fall protection on ladders when additional significant hazards such as impalement, rotating machinery, or electrical shock are present.

- Ladders shall be maintained free of oil, grease, and other slipping hazards.
- Workers shall maintain their body's center of gravity between the side rails at all times while working from a ladder. In addition, workers shall avoid work from a ladder that involves significant pushing, pulling, or any action that may dislodge the person from the ladder.
- The top two steps of a step ladder shall not be used as steps. This requirement does not apply to step ladders with three steps or less or to step ladders with a guard rail-equipped work platform at the top.

18.0 SCAFFOLDING:

- Scaffolds or elevated platforms shall be constructed, maintained, and used in accordance with the applicable regulations and company standards.
- Contractor shall ensure that a Competent Person is assigned to supervise scaffold erection, dismantling, alteration, and movement.
- Contractor shall ensure that all scaffold materials and planking are thoroughly inspected for defects prior to use.
- Where there is a hazard to workers working below an elevated scaffold, toe boards shall be in place.
- Climbing or working from the handrail, mid-rail, or brace members of the scaffolding is prohibited.

19.0 FALL PROTECTION

- Definitions:
 - a) Fall Protection Competent Person – A person who is knowledgeable of:
 - b) The fall hazards at the worksite;
 - c) Correct procedures for assembling, maintaining, disassembling, and inspecting fall arrest equipment; and
 - d) The operation of guardrail systems, Personal Fall Arrest Systems (defined below), warning line systems, safety monitoring systems, and other protection to be used.
 - e) Personal Fall Arrest System – A system used to arrest a person in a fall from a working level at height. It consists of an anchorage, connectors, and body harness. It may include a lanyard, deceleration device, lifeline, or combination of these.
- A Personal Fall Arrest System shall be worn:
 - a) On work surfaces where potential drop is greater than 6 feet (1.8 meters).
 - b) When working on unfinished structures greater than 6 feet (1.8 meters) in height where the work surface is without guardrails, toe boards, or gated access ladders;

c) When working on areas within 6 feet (1.8 meters) of the edge of a work surface greater than 6 feet in height or within 6 feet of any unguarded opening, skylight, service duct, stairwell, or elevator shaft on a roof or unfinished level of a structure;

d) When working along unguarded locations at the edge of a well, pit, shaft, excavation, trench, or similar location 6 feet or more in depth when the excavation is not readily seen because of plant growth or other visual barrier;

e) Whenever deemed necessary by a safety officer.

- Contractor shall inspect all components of a Personal Fall Protection System before each use and replace if necessary. Fall protection equipment that has been subjected to impact loading shall be immediately removed from service and shall be inspected by the manufacturer before reuse or destroyed and replaced.
- Contractor shall ensure that components of a Personal Protection System are free from defect such as cuts, tears, abrasions, mold, undue stretching, missing or degraded stitching, alterations, or additions that might affect its efficiency. Contractor shall also inspect for damage due to chemical exposure, deterioration, distorted hooks, or faulty hook springs, loose or damaged mountings, non-functioning parts, wearing, or internal deterioration in the ropes or webbing.
- Contractor shall follow the manufacturer's recommended procedures for fitting, adjusting, using, inspecting, testing, and caring for fall protection equipment.
- A Personal Fall Arrest System shall not be used as a primary suspension device for positioning, or as a retrieval system, or for transporting materials.
- Contractor shall evaluate the compatibility of all fall arrest systems and anchorage points prior to use.
- Contractor shall calculate tie-off distances accurately to limit a fall to a maximum of 6 feet, considering lanyard elongation, Work position, proximity to fall area, and the location of fall hazards. The anchorage and tie-off points should be located to avoid obstructions in the potential

20.0 CHEMICAL SAFETY

- JPL uses a variety of chemicals that Contractors may encounter. JPL maintains MSDS's for these chemicals. Contractors may request a copy of any/all MSDS's for chemicals to which they are (or may be) exposed by contacting their JPL contact.
- Use of JPL chemicals by a Contractor for any purpose must be authorized in advance by the JPL contact.
- A current MSDS must be readily available and maintained by the Contractor for all chemicals brought onto JPL premises.

21.0 HAND TOOLS

- All hand tools, including portable electrical tools, and other equipment brought onto JPL premises must be in safe condition. JPL reserves the right to prohibit the use of defective tools, ladders, etc. and dictate the removal of defective equipment.

22.0 CRANES AND HOISTS

- JPL owned Overhead Cranes and Hoists are not to be operated by Contractors unless they receive advance authorization from the JPL contact.
- Upon authorization, only Contractor Employees for whom the Contractor has provided training sufficient to meet the standards for cranes and hoists (including applicable licensing) will be allowed to operate this equipment.
- Contractor owned Cranes and Hoists must be operated under the requirements of Safety Standards for Overhead and Gantry Cranes and Monorail Systems. This includes training/licensing requirements, equipment inspection, and safe equipment operation requirements.
- Any crane, hoist or rigging system that is not safe to operate must be tagged out of service and not operated until repaired/serviced.

23.0 HOUSEKEEPING

- Job site housekeeping is the contractor's responsibility and the job site must be as clean and orderly as possible while work is being performed. Good housekeeping practices are of utmost importance in preventing injuries.
- DO NOT LITTER. Contractors are responsible to keep the work area(s) in a neat and orderly condition at all times. All material must be cleaned up as the job progresses.
- All roadways, passageways, and operating areas must be kept clear at all times

24.0 WASTE DISPOSAL

- Proper identification of waste generated during work on-site is critical. All wastes must be segregated and managed according to applicable regulatory requirements.
- The Contractor is responsible for the removal of any waste generated.
- It is the responsibility of the Contractor to ensure proper waste management practices while performing services for JPL. Prior to any work, the Contractor will assess what wastes will be generated and communicate to the JPL contact any hazardous, non-hazardous, universal or construction wastes that will be generated and the methods that will be used to collect, manage, and dispose of these wastes.
- Discharge of any material onto the ground is strictly prohibited by State/National Laws and JPL HSE policy. If any spill/discharge occurs, report it immediately to your JPL Contact (i.e. leak from truck of oil, gas, or product being transported).
- Any questions regarding proper classification and/or disposal of wastes generated must to be brought to the attention of your JPL contact.

25.0 ROOF WORK

- The portion of a roof where Contractors are working must be kept free of slippery conditions.
- All roof work performed on JPL premises must be conducted in accordance with Safety Standard for Fall Protection.

26.0 RIGGING

- All rigging must be done in accordance with applicable regulations.

27.0 OVERHEAD WORK

- Contractor shall ensure that workers are trained to recognize the hazards of working around overhead utility lines and how to minimize the chance of contact.
- Contractor shall take precautions to ensure the safety of workers and ensure the integrity of the existing overhead utility lines.
- Contractor shall conduct a hazard assessment to identify and mitigate hazards prior to working around overhead utility lines. The hazard assessment shall include the following:
 - Identifying all overhead utility lines (on or off the worksite) that may be impacted by the Work;
 - Verifying that appropriate signage and visual warning devices are installed to alert workers to the hazards;
 - Clearly marking or otherwise restraining all lifting or boom-type equipment to show the maximum height or extension possible as measured from ground level or to limit the maximum limit of extension, respectively; and
 - Using adequately trained and dedicated spotters at locations where equipment and vehicles pass or work under or around utility lines.
- Physical barriers: Non-conductive, highly visible devices (e.g., goal posts, barricade tape) set outside the limits of approach (limits shall vary by jurisdiction, land restriction, and voltages) on both the coming and going away sides.
- Site-specific controls prescribed and authorized for use by the utility owner (e.g., shielding, de-energizing, bonding, insulating).
- Keep all equipment attachments in the lowest possible position when traveling under overhead utilities.
- Use dry tag lines made of a nonconductive type material when working near energized lines.
- All ladders used around power lines shall be made of non-conductive materials;
- Use only non-conductive chokers, slings, and lifting devices during material handling activities;
- Keep materials bonded at all times when transporting conductive loads, (e.g., pipe, air compressor, pumps) in the proximity of high voltage lines.

- All overhead work must be conspicuously barricaded or otherwise marked to prevent anyone from walking or driving under the work area.
- Overhead work creating sparks requires a Hot Work Permit.
- Overhead work creating falling debris requires additional protection for personnel and equipment that may be affected by the falling debris
- All scaffolds must have full flooring within the frame, with cleats, toe boards, and railings and meet BIS requirements.

28.0 ELEVATED WORK

- General: When working overhead, Contractor shall protect people below. Contractor shall ensure that tools, materials, and equipment subject to falling from height are adequately secured before Work is performed. Tools and materials shall be handed up or down, but never thrown. When it is necessary to hoist tools with a rope, exercise care to ensure the tools are securely attached to the line or loaded into a container and there is no danger of them being dropped.

29.0 WORK ZONE TRAFFIC CONTROL

- If a Work activity is conducted on or near a road, Contractor shall comply with all applicable regulations.
- Contractor shall provide all signs, barriers, flaggers, and other notification necessary to protect its workers and the public from damage, injury, or loss. Barricades at public areas (e.g., road crossings) shall have flashing lights/ reflective during hours of darkness.
- All work conducted in on or near a road at night requires the use of high visibility traffic vests.

30.0 PENALTY SYSTEM

- On non-compliance of PPEs and other safety instructions, following penalties will be imposed on the contracting agency as per below mentioned violations. The amount towards the violation shall be deducted from monthly bill of the contracting agency.

S.No.	Violation	Penalties
1	Non-use of any of PPE like Safety Helmet with chin strap, Safety shoes by individuals	Lapses & non-compliance in safety related issues in repetitive manner will be viewed very seriously by JPL Management and further a monetary amount as deemed fit by JPL's Management will be debited from your Bill without any consideration.
2	Non-use of Full body Harness at height	
3	Repetition of violation by employees of same contractor within a week	
4.	Over speeding or wrong side driving	
5.	Non reporting of accident	
6.	Working without work permit	

7.	Non-use of proper tools & tackles i.e. Cutting torch without Flash back arrestor at both ends, grinder without Guard, Lifting tools and tackles without certification etc.	
8	Allow to work on visitor pass	
9	Overloading during material handling	
10	Smoking at workplace	
11	Non-use of PPE's as per the job requirement.	

IN CASE AGENCY OR INDIVIDUAL IS A FREQUENT DEFAULTER, A RED MARK SHALL BE RECORDED IN HIS PERFORMANCE RECORD & THE AGENCY SHALL BE BARRED AND BLACK LISTED FOR REPETITIVE NON COMPLIANCES ON FRONT OF SAFETY.

31.0 ACCIDENT REPORTING AND INVESTIGATION

- The Contractor must immediately report any injury, illness, or near miss that occurs at JPL premises to the JPL contact.
- The information related to the incident must be provided to JPL contact as per the JPL Incident reporting and investigation Procedure.

32.0 GENERAL RULES TO FOLLOW

- Always store materials in a safe manner. Tie down or support piles if necessary to prevent falling, rolling, or shifting.
- Dust scraps, oil or grease should not be allowed to accumulate. Good housekeeping is a part of the job.
- Trash piles must be removed as soon as possible. Trash is a safety and fire hazard.
- Remove or bend over the nails in lumber that has been used or removed from a structure.
- Immediately remove all loose materials from stairs, walkways, ramps, platforms, etc.
- Do not block aisles, traffic lanes, fire exits, gangways, or stairs.
- Avoid shortcuts – use ramps, stairs, walkways, ladders, etc.
- Standard guardrails must be erected around all floor openings and excavations must be barricaded. Contact your supervisor for the correct specifications.
- Do not remove, deface or destroy any warning, danger sign, or barricade, or interfere with any form of protective device or practice provided for your use or that is being used by other workers.
- Get help with heavy or bulky materials to avoid injury to yourself or damage to material.
- Keep all tools away from the edges of scaffolding, platforms, shaft openings, etc.
- Do not use tools with split, broken, or loose handles, or burred or mushroomed heads. Keep cutting tools sharp and carry all tools in a container.
- Know the correct use of hand and power tools. Use the right tool for the job.
- Know the location and use of fire extinguishing equipment and the procedure for sounding a fire alarm.
- Proper guards or shields must be installed on all power tools before use. Do not use any tools without the guards in their proper working condition. No “homemade” handles or extensions (cheaters) will be used!
- All electrical power tools (unless double insulated), extension cords, and equipment must be properly

grounded.

- All electrical power tools and extension cords must be properly insulated. Damaged cords must be replaced.
- Do not operate any power tool or equipment unless you are trained in its operation and authorized by your firm to do so.
- All electrical power equipment and tools must be grounded or double insulated.
- Use tools only for their designed purpose.